



Cboe Titanium U.S. Options Complex Multicast PITCH Specification

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Introduction

Note that this specification will be the standard specification to be used for complex options on all Cboe-affiliated Options Exchange platforms.

Cboe customers may use Complex Multicast PITCH to receive real-time depth of book quotations and execution information direct from Cboe. This feed will only include quotations and executions related to complex orders.

Complex Multicast PITCH cannot be used to enter orders. For order entry, refer to the appropriate US Options FIX or BOE Specifications.

A Gig-Shaped version of the Complex Multicast PITCH feed is available from both of Cboe's datacenters. Customers may choose to take one or more of the following Complex Multicast PITCH feed options depending on their location and connectivity to Cboe.

Complex Multicast PITCH Feed Descriptions

Table 1. Complex Multicast PITCH Feed Descriptions

EXCHANGE	SHAPING (GIG)	SERVED FROM DATA CENTER (PRIMARY/SECONDARY)	MULTICAST FEED ID
C1 Options	Gig	Primary	CAC
C1 Options	Gig	Primary	CBC
C1 Options	Gig	Secondary	CEC
C2 Options	Gig	Primary	WAC
C2 Options	Gig	Primary	WBC
C2 Options	Gig	Secondary	WEC
EDGX Options	Gig	Primary	EAC
EDGX Options	Gig	Primary	EBC
EDGX Options	Gig	Secondary	EEC
BZX Options	Gig	Primary	OAC
BZX Options	Gig	Primary	OBC
BZX Options	Gig	Secondary	OEC

24x5 Feed Hours and System Restart (C1 Only)

For C1 Options operating in 24x5 mode, the PITCH feed starts on Sunday at approximately 1:00 p.m. ET and shuts down on Friday at approximately 5:30 p.m. ET. A daily restart occurs between 5:30 and 7:00 p.m. ET each day at which time sequences will be reset. The daily restart is typically observed between 5:30 p.m. and 6:00 p.m. ET, but could occur later if needed for operational reasons. Feed startup and shutdown times may be adjusted without notice.

Under normal operations, it is expected that the order books are cleared (Delete Order messages sent for any open orders, including GTC and GTD orders), prior to the daily restart and reset of sequences. Persisted GTC and GTD orders will be added back onto the order books starting at approximately 10 minutes prior to the scheduled queuing time for each underlying symbol.

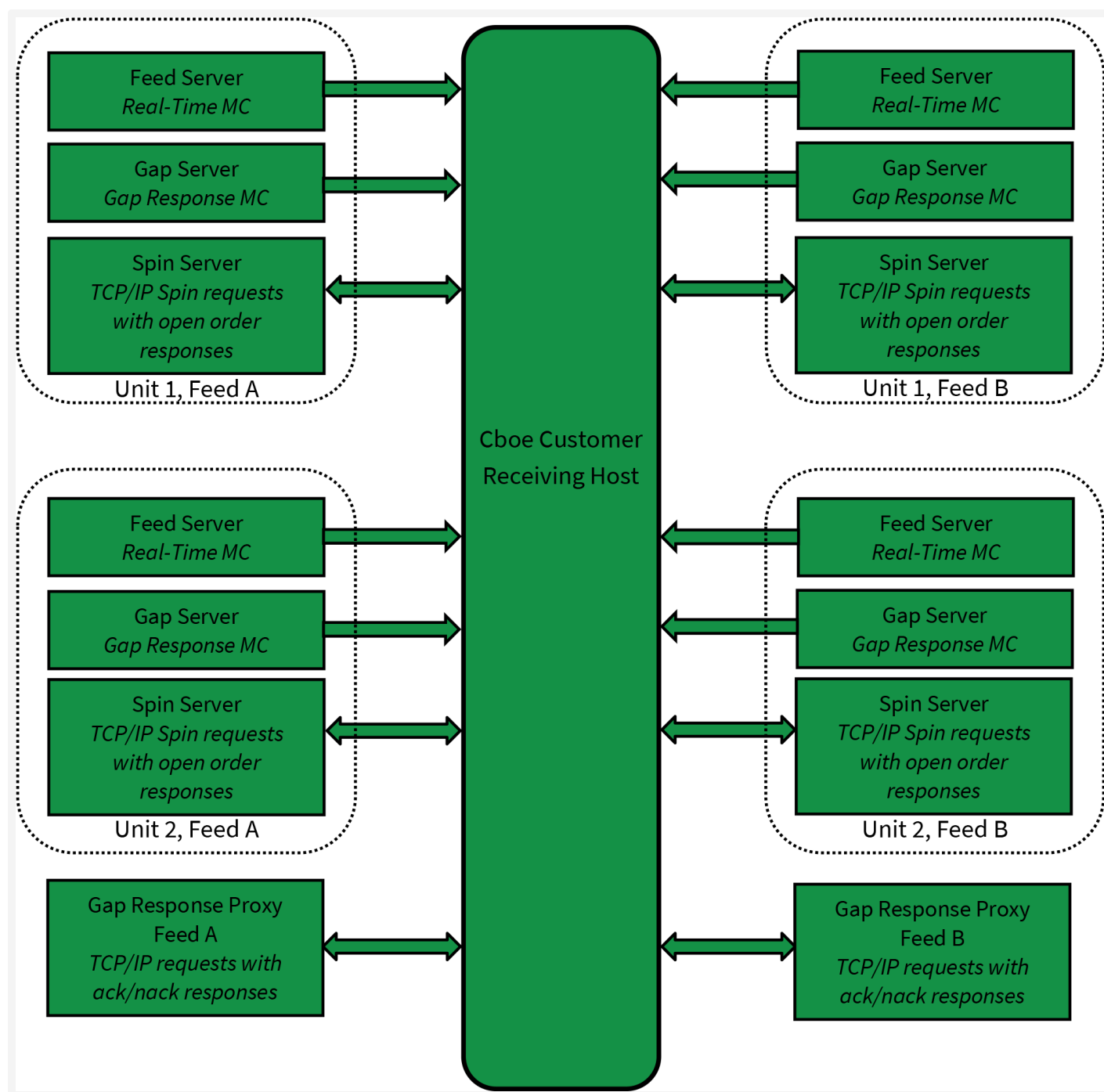
Feed Connectivity Requirements

Gig Shaped feeds are available to customers with a minimum of 1 Gb/s of connectivity to Cboe via cross connect or dedicated circuit.

Customers with sufficient connectivity may choose to take more than one Gig-Shaped feed from the Cboe datacenters and arbitrate the feeds to recover lost data. It should be noted that feeds from the secondary datacenter will have additional latency for those co-located with Cboe in the primary datacenter due to proximity.

Cboe Complex Multicast PITCH real-time events are delivered using a published range of multicast addresses divided by symbol range units. Dropped messages can be requested using a TCP/IP connection to one of Cboe's Gap Request Proxy (GRP) servers with replayed messages being delivered on a separate set of multicast ranges reserved for packet retransmission. Intraday, a spin of all open orders may be requested from a Spin Server. This allows a client to become current without requesting a gap for all messages up to that point in the day.

The following diagram is a logical representation Complex Multicast PITCH feed message flow between Cboe and a customer feed handler that is listening to the "A" and "B" instances of two units:



Symbol Ranges, Units, and Sequence Numbers

Symbols will be separated by underlying into units by a published distribution. Symbol distribution will not change intra-day. Cboe does, however, **reserve the right to add multicast addresses or change the symbol distribution** with prior notice to customers. Care should be taken to ensure that address changes, address additions, and symbol distribution changes can be supported easily.

Message sequence numbers are incremented by one for every sequenced message within a particular symbol unit. It is important to understand that one or more units will be delivered on a single multicast address. As with symbol ranges, unit distribution across multicast addresses will not change intra-day, but may change after notice has been given.

Symbol distribution across units as well as unit distribution across multicast addresses are identical for real-time and gap response multicast addresses.

Complex Options Specific Symbol Processing

Cboe has implemented a Complex Instrument Creation (CIC) process due to the seemingly infinite number of combinations that can make up a complex instrument. This allows the Complex Multicast PITCH specification to be consistent with the equities, standard and auction options Multicast PITCH specifications. This CIC process significantly reduces the size of the Complex Multicast PITCH feed and allows customers to use the same feed handler for Cboe equity, options, and futures exchanges.

Real-time CIC messages are available on each unit's multicast feed. **Complex Instrument Definition Expanded** messages are used to map the 6 character feed Complex Instrument ID (CID) to the complex instrument definition. A complex instrument definition consists of two or more option legs. **The complex instrument is valid only for the current trading date on which it was created.** Once a complex instrument is created, it cannot be deleted or modified for the remainder of the trading day.

Complex instruments can be created from pre-market through the end of trading. When a complex instrument is created, a sequenced **Complex Instrument Definition Expanded** message is disseminated at the time of creation and is sent in a background loop as an unsequenced message for the remainder of the trading day as bandwidth allows.

Gap Request Proxy and Message Retransmission

Requesting delivery of missed data is achieved by connecting to the Cboe Gap Request Proxy (GRP) for the complex options data feed. Customers who do not wish to request missed messages do not need to connect to a GRP for any reason or listen to the multicast addresses reserved for message retransmission. Customers choosing to request missed data will need to connect to their assigned GRP, log in, and request gap ranges as necessary. All gap requests will be responded to with a **Gap Response** message. A **Gap Response Status** code of Accepted signals that the replayed messages will be delivered via the appropriate gap response multicast address. Any other **Gap Response Status** code will indicate the reason that the request cannot be serviced.

Gap requests are limited in message count, frequency, and age by the GRP. Gap requests will only be serviced if they are within a defined sequence range of the current multicast sequence number for the requested unit. Customers will receive a total daily allowance of gap requested messages. In addition, each customer is given renewable one second and one minute gap request limits.

If more than one gap request is received for a particular unit/sequence/count combination within a short timeframe, all requests will receive a successful **Gap Response** message from the GRP, but only a single replayed message will be sent on the gap response multicast address.

If overlapping gap requests are received within a short period of time, the gap server will only send the union of the sequence ranges across grouped gap requests. Customers will receive gap responses for their requested unit/sequence/count, but receivers should be prepared for the **gap responses to be delivered via multicast in non-contiguous blocks**.

Gap acknowledgments or rejects will be delivered to users for every gap request received by the GRP. Users should be prepared to see replayed multicast data before or after the receipt of the gap response acknowledgment from the GRP.

Spin Servers

A Spin Server is available for each unit. The server allows customers to connect via TCP and receive a spin of all complex instrument definitions and currently open orders with limited trading conditions on that unit. By using the spin, a customer can get the current complex book quickly in the middle of the trading session without worry of gap request limits. The Spin Server for each unit listens on its own address and/or TCP port.

Upon successful login and periodically thereafter, a **Spin Image Available** message is sent which contains a sequence number indicating the most recent message applied to the complex book. Using a **Spin Request** message, a customer may request a spin for the orders up to a sequence number noted within one of the last ten **Spin Image Available** messages distributed. If the **Spin Request** submitted does not present a sequence number that matches one of the last ten **Spin Image Available** messages distributed, the spin will return orders up to the next closest sequence number reported through a **Spin Image Available** message that is greater than the sequence number requested.

In the case a customer sends a sequence number in a **Spin Request** that is higher than the sequence number reported by the most recent **Spin Image Available** message, the next spin image to be generated will be returned when it is available. If the requested sequence number is still higher at that time, an O (Out of Range) error will be generated.

A spin consists only of **Complex Instrument Definition Expanded**, **Exchange Designated Complex Instrument Definition**, **Add Order (long and/or short)**, **Trading Status** and **Time** messages. **Trading Status** messages will be sent in spins for all complex instruments that are not S=Suspended, which results in at least two messages for every complex instrument that has not been Suspended since system startup. A **Time** message will be sent as the last message in a spin if the last **Time** message sent on a spin is older than the last received time from the internal market data producers.

Spins will not contain any message for an order which is no longer on the book. While receiving the spin, the customer must buffer multicast messages received. If the **Spin Image Available** message sequence number is the customer's reference point, multicast messages with larger sequence numbers should be buffered. If a non- **Spin Image Available** sequence number is the customer's reference point from which they send in their **Spin Request**, they should buffer from that point on, but note that the spin they will receive will contain sequence numbers beyond that point which may be disregarded. When a **Spin Finished** message is received, the buffered messages must be applied to spun copy of the book to bring it current.

Customers can also use the Spin Server to request a spin of all **Symbol Mapping**, **Complex Instrument Definition Expanded** and **Exchange Designated Complex Instrument Definition** messages by sending an **Instrument Definition Request**. The Spin Server can only process one spin at a time. Customers will need to wait for a **Spin Finished** or **Instrument Definition Finished** message before submitting another request.

[Spin Messages](#) on page 0 shows an example flow of messages between a customer and Cboe's Multicast PITCH feed and Spin Server.

Protocol

Cboe users may use the PITCH 2.X protocol over multicast to receive real-time full depth of complex book quotations and execution information direct from Cboe.

Message Format

The messages that make up the PITCH 2.X protocol are delivered using **Sequenced Unit Header** which handles sequencing and delivery integrity. All messages delivered via multicast as well as to/from the Gap Request Proxy (GRP) will use the **Sequenced Unit Header** for handling message integrity.

All UDP delivered events will be self-contained. Developers can assume that UDP delivered data will not cross frame boundaries and a single Ethernet frame will contain only one **Sequenced Unit Header** with associated data.

TCP/IP delivered events from the GRP may cross frames as the data will be delivered as a stream of data with the TCP/IP stack controlling Ethernet framing.

The PITCH data feed is comprised of a series of dynamic length sequenced messages. Each message begins with *Length* and *Message Type* fields. Cboe reserves the right to add message types and grow the length of any message without notice. Customers should develop their decoders to deal with unknown message types and messages that grow beyond the expected length. Messages will only be grown to add additional data to the end of a message.

Data Types

The following field types are used within the **Sequenced Unit Header**, GRP messages, and PITCH 2.X.

- **Alphanumeric** fields are left justified ASCII fields and space padded on the right.
- **Binary** fields are unsigned and sized to Length bytes and ordered using Little Endian convention (least significant byte first).
- **Signed Binary** fields are signed and sized to Length bytes and ordered using Little Endian convention (least significant byte first).
- **Binary Signed Short Price** fields are signed Little Endian encoded 2 byte binary fields with 2 implied decimal places (denominator = 100). The short price range is -327.68 to +327.67. Prices outside of this range will use the long price.
- **Binary Signed Long Price** fields are signed Little Endian encoded 8 byte binary fields with 4 implied decimal places (denominator = 10,000).
- **Bit Field** fields are fixed width fields with each bit representing a Boolean flag (the 0 bit is the lowest significant bit; the 7 bit is the highest significant bit).
- **Printable ASCII** fields are left justified ASCII fields that are space padded on the right that may include ASCII values in the range of 0x20 - 0x7e.
- **Binary Date** fields are 4 byte unsigned Little Endian values where the base-10 representation is the YYYYMMDD representation of that date. For example, October 30, 2023 would be represented as 20,231,030 (20231030).
- **Time Offset** fields are 4 byte unsigned Little Endian values that represent the number of nanoseconds since the last **Time** message. These fields are populated to nanosecond precision. For example, a nano time of 1,737,580,407,123,456,789 ("Wednesday, January 22, 2025 9:13:27.123456789") would return value = 123456789.

Message Framing

Depth of book update messages will be combined into single UDP frame where possible to decrease message overhead and total bandwidth. The count of messages in a UDP frame will be communicated using the **Sequenced Unit Header**. Framing will be determined by the server for each unit and site. The content of the multicast across feeds (e.g. A/B & Gig-Shaped) will be identical, **but framing will not be consistent across feeds**. Receiving processes that receive and arbitrate multiple feeds cannot use frame level arbitration to fill gaps.

Sequenced Unit Header

The **Sequenced Unit Header** is used for all Cboe Complex Multicast PITCH messages as well as messages to and from the Gap Request Proxy (GRP) and Spin Servers.

Sequenced and un-sequenced data may be delivered using the **Sequenced Unit Header**. Un-sequenced headers will have a 0 value for the sequence field and potentially for the unit field. All messages sent to and from the GRP and Spin Server are un-sequenced while multicast may contain sequenced and un-sequenced messages.

Sequenced messages have implied sequences with the first message having the sequence number contained in the header. Each subsequent message will have an implied sequence one greater than the previous message up to a maximum of count messages. Multiple messages can follow a **Sequenced Unit Header**, but a combination of sequenced and un-sequenced messages cannot be sent with one header.

The sequence number for the first message in the next frame can be calculated by adding the *Hdr Count* field to the *Hdr Sequence*. This technique will work for sequenced messages and heartbeats.

Table 2. Sequenced Unit Header

FIELD	OFFSET	LENGTH	VALUE/TYPE	DESCRIPTION
<i>Hdr Length</i>	0	2	Binary	Length of entire block of messages. Includes this header and <i>Hdr Count</i> messages to follow.
<i>Hdr Count</i>	2	1	Binary	Number of messages to follow this header.
<i>Hdr Unit</i>	3	1	Binary	Unit that applies to messages included in this header.
<i>Hdr Sequence</i>	4	4	Binary	Sequence of first message to follow this header.
Total Length = 8 bytes				

Heartbeat Messages

The **Sequenced Unit Header** with a count field set to 0 will be used for **Heartbeat** messages. During trading hours **Heartbeat** messages will be sent from the GRP and all multicast addresses if no data has been delivered within 1 second. **Heartbeat** messages never increment the sequence number for a unit, but can be used to detect gaps on the real-time multicast channels during low update rate periods.

Heartbeat messages on the real-time multicast addresses during trading hours will have a *Hdr Sequence* value equal to the sequence of the next sequenced message to be sent for the unit.

Heartbeats on gap multicast addresses will always have the *Hdr Sequence* field set to 0. All **Heartbeat** messages sent to and from the GRP are considered un-sequenced and should have sequence and unit fields set to 0.

Outside of trading hours Cboe sends **Heartbeat** messages on all real-time and gap channels with a sequence of 0 to help users validate multicast connectivity. **Heartbeat** messages may not be sent from 12:00 a.m. - 1:00 a.m. ET or during maintenance windows.

Cboe expects **Heartbeat** messages to be sent to the GRP and Spin Servers on live connections no less than every 5 seconds. Failure to receive 2 consecutive **Heartbeat** messages will result in the GRP or Spin Servers terminating the client connection.

PITCH 2.X Messages

With the exception of **Time Reference** and **Time** messages, each PITCH message reflects the order addition, order deletion, order modification or execution of an order in the system.

Time Reference Message Fields (C1 Only)

The **Time Reference** message is used to provide a midnight reference point for recipients of the feed. It is sent whenever the system starts up and when the system crosses a midnight boundary. All subsequent **Time** messages for the same unit will use the last *Midnight Reference* until another **Time Reference** message is received for that unit. The **Time Reference** message includes the *Trade Date*, so most other sequenced messages will not include that information.

Time Reference messages will be included in a spin response.

Table 3. Time Reference

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field.
<i>Message Type</i>	1	1	0xB1	Time Reference Message
<i>Midnight Reference</i>	2	4	Binary	Midnight Eastern Time reference time for subsequent Time messages, expressed as number of whole seconds since the Epoch (Midnight January 1, 1970 UTC).
<i>Time</i>	6	4	Binary	Number of whole seconds elapsed since the start of the current Eastern Time calendar day, derived by converting the current Eastern wall clock time (HH:MM:SS) to seconds. On Daylight Saving Time transition days, <i>Time</i> reflects wall clock seconds and does not adjust for the UTC offset change. On a DST spring-forward day, the maximum value will be 82,799 (23 hours). On a DST fall-back day, values between 3,600 and 7,199 will appear twice. Recipients requiring unambiguous absolute timestamps should use <i>Epoch Time</i> (where available) or <i>Midnight Reference</i> in the Time Reference message.
<i>Time Offset</i>	10	4	Time Offset	Nanosecond offset from last unit timestamp.
<i>Trade Date</i>	14	4	Binary Date	Current Trade Date
Total Length = 18 bytes				

Time Message Fields

A **Time** message is immediately generated and sent when there is a PITCH event for a given clock second. If there is no PITCH event for a given clock second, then no **Time** message is sent for that second. The *Time* field is the number of seconds relative to midnight Eastern Time. All subsequent time offset fields for the same unit will use the new **Time** value as the base until another **Time** message is received for the same unit. On **C1 only**, the **Time** message will also include the *Epoch Time* field, which is the current time represented as the number of whole seconds since the Epoch (midnight January 1, 1970).

For **C1 only**, a given trading day may span multiple calendar days. C1 options market data recipients must prepare for a crossing of the midnight ET boundary. At such time, a new **Time Reference** message will be sent and the *Time* field in subsequent **Time** messages will reset to reflect the number of seconds from the most recent midnight ET time.

Table 4. Time

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field.
<i>Message Type</i>	1	1	0x20	Time Message
<i>Time</i>	2	4	Binary	Number of whole seconds elapsed since the start of the current Eastern Time calendar day, derived by converting the current Eastern wall clock time (HH:MM:SS) to seconds. On Daylight Saving Time transition days, <i>Time</i> reflects wall clock seconds and does not adjust for the UTC offset change. On a DST spring-forward day, the maximum value will be 82,799 (23 hours). On a DST fall-back day, values between 3,600 and 7,199 will appear twice. Recipients requiring unambiguous absolute timestamps should use <i>Epoch Time</i> (where available) or <i>Midnight Reference</i> in the Time Reference message.
<i>Epoch Time</i>	6	4	Binary	C1 Only Number of whole seconds since the Epoch (midnight January 1, 1970 UTC).
Total Length = 6 bytes, 10 bytes on C1 Only				

Unit Clear Message Fields

The **Unit Clear** message instructs feed recipients to clear all orders for the Cboe complex book in the unit specified in the **Sequenced Unit Header**. For **Equities only**, this message will be sent at startup each day. It would also be distributed in certain recovery events such as a data center fail-over.

Table 5. Unit Clear

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x97	Unit Clear Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
Total Length = 6 bytes				

Transaction Begin Message Fields

The **Transaction Begin** message indicates any subsequent messages, up to the accompanying **Transaction End** message, are all part of the same transaction block. All PITCH messages corresponding to such an event would be included between a **Transaction Begin** and **Transaction End**. It is important to note that any PITCH Message Type may be included in a transaction block and there is no guarantee that the messages apply to the same price level or even the same Symbol. **Transaction Begin** messages do not alter the book and can be ignored if messages are being used solely to build a book.

Feed processors can use a transaction block as a trigger to postpone publishing a quote update until the end of the transaction block.

Table 6. Transaction Begin

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field.
<i>Message Type</i>	1	1	0xBC	Transaction Begin Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp.
Total Length = 6 bytes				

Transaction End Message Fields

The **Transaction End** message indicates that a transaction indicated by a previous **Transaction Begin** message has completed. **Transaction End** messages do not alter the book and can be ignored if messages are being used solely to build a book.

Table 7. Transaction End

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field.
<i>Message Type</i>	1	1	0xBD	Transaction End Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp.
Total Length = 6 bytes				

Complex Instrument Definition Expanded Message Fields

A **Complex Instrument Definition Expanded** message represents a complex instrument that is available to place orders. It is sent as a sequenced message the first time a **Complex Instrument Definition Expanded** message is sent for a symbol. These messages will also be sent continuously through the day as an unsequenced message (sequence = 0) at variable rates as bandwidth allows. The *Time offset* field should be ignored on an unsequenced **Complex Instrument Definition Expanded** message.

The **Complex Instrument Definition Expanded** message will contain two or more repeating groups of leg definitions. There is a limit of 16 leg definitions, one of which may be an equity leg.

Table 8. Complex Instrument Definition Expanded

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field.
<i>Message Type</i>	1	1	0x9A	Complex Instrument Definition Expanded Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp.
<i>Complex Instrument Id</i>	6	6	Printable ASCII	Complex Instrument Id right padded with spaces.
<i>Complex Instrument Underlying</i>	12	8	Printable ASCII	Complex Instrument Underlying right padded with spaces.
<i>Complex Instrument Type</i>	20	4	Alphanumeric	4 character field; each field describes a characteristic. Character 1: Complex Option Type O = All legs are options E = One leg is an equity leg Characters 2-4: Reserved
<i>Leg Count</i>	24	1	Binary	The number of legs in the complex instrument. The maximum number of legs is 16.
The following fields repeat <i>Leg Count</i> times for multi-leg strategies. <i>Leg Index</i> is zero-based.				
<i>Leg Symbol</i>	25 + Leg Index * 13	8	Printable ASCII	Option or Equity Symbol of leg, right padded with spaces.
<i>Leg Ratio</i>	33 + Leg Index * 13	4	Signed Binary	Leg ratio (positive for buy-side, negative for sell-side). For options this is the number of contracts, for equities this is the number of shares.
<i>Leg Security Type</i>	37 + Leg Index * 13	1	Alphanumeric	O = Leg is an Option instrument E = Leg is an Equity instrument
Total Length = 25 + (Leg Count * 13) bytes				

Exchange Designated Complex Instrument Definition Message Fields (C1 Only)

The **Exchange Designated Complex Instrument Definition** (EDCID) message represents supplemental information associated with an exchange-designated complex instrument and is sent as a sequenced message the first time it is sent. These messages are also sent continuously throughout the day as an unsequenced message (sequence = 0) at variable rates as bandwidth allows. The *Time offset* field should be ignored on unsequenced EDCID messages.

This message is sent in addition to a **Complex Instrument Definition Expanded** message for each exchange-designated complex instrument. All complex instruments, including exchange designated complex instruments, will be represented with standard **Complex Instrument Definition Expanded** messages. The purpose of the EDCID messages is to associate complex instruments with dependent Cboe products and services through the EDCI Type and Subtype fields.

Note that the same complex instrument can be associated with more than one dependent product, resulting in more than one EDCID message sharing the same *Complex Instrument Id* field value.

The EDCID message contains two or more repeating groups of leg definitions. There is a limit of 16 leg definitions.

Table 9. Exchange Designated Complex Instrument Definition

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field.
<i>Message Type</i>	1	1	0x9F	Exchange Designated Complex Instrument Definition message.
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp.
<i>Complex Instrument Id</i>	6	6	Printable ASCII	Complex Instrument Id right padded with spaces.
<i>Complex Instrument Underlying</i>	12	8	Printable ASCII	Complex Instrument Underlying right padded with spaces.
<i>EDCI Type</i>	20	20	Printable ASCII	Exchange Designated Complex Instrument type right padded with spaces.
<i>EDCI Subtype</i>	40	20	Printable ASCII	Exchange Designated Complex Instrument subtype right padded with spaces.
<i>Reserved</i>	60	2	Reserved	Reserved
<i>Leg Count</i>	62	1	Binary	The number of legs in the complex instrument. The maximum number of legs is 16.
The following fields repeat <i>Leg Count</i> times for multi-leg strategies. <i>Leg Index</i> is zero-based.				
<i>Leg Symbol</i>	63 + Leg Index * 10	6	Printable ASCII	Option symbol of leg, right padded with spaces.
<i>Leg Ratio</i>	69 + Leg Index * 10	4	Signed Binary	Leg ratio (positive for buy-side, negative for sell-side).
Total Length = 63 + (Leg Count * 10) bytes				

Symbol Mapping Message Fields

A **Symbol Mapping** message is used to map the 6 character multicast feed symbol field to an OSI symbol and Underlying. These messages are not sequenced (sequence = 0) and are sent continuously through the day at variable rates as bandwidth allows.

Table 10. Symbol Mapping

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field.
<i>Message Type</i>	1	1	0x2E	Symbol Mapping Message.
<i>Feed Symbol</i>	2	6	Printable ASCII	<i>Symbol</i> right padded with spaces.
<i>OSI Symbol</i>	8	21	Printable ASCII	OSI Symbol
<i>Symbol Condition</i>	29	1	Alphanumeric	N = Normal C = Closing Only
<i>Underlying</i>	30	8	Alphanumeric	Symbol of underlying equity right padded with spaces. All spaces if not available or not applicable.
Total Length = 38 bytes				

Add Order Message Fields

An **Add Order** message represents a newly accepted visible order on the Cboe complex book. It includes a day-specific Order Id assigned by Cboe to the complex order.

Table 11. Add Order (long)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x21	Add Order Message (long)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	Day-specific identifier assigned to this order
<i>Side Indicator</i>	14	1	Alphanumeric	B = Buy Order S = Sell Order
<i>Quantity</i>	15	4	Binary	Instrument quantity added to the complex book (may be less than the number entered).
<i>Complex Instrument Id</i>	19	6	Printable ASCII	<i>Complex Instrument Id</i> right padded with spaces.
<i>Price</i>	25	8	Binary Signed Long Price	The limit order price
<i>Reserved</i>	33	1	Reserved	Reserved
Total Length = 34 bytes				

Table 12. Add Order (short)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x22	Add Order Message (short)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	Day-specific identifier assigned to this order
<i>Side Indicator</i>	14	1	Alphanumeric	B = Buy Order S = Sell Order
<i>Quantity</i>	15	2	Binary	Instrument quantity being added to the complex book (may be less than the number entered).
<i>Complex Instrument Id</i>	17	6	Printable ASCII	Complex Instrument Id right padded with spaces.
<i>Price</i>	23	2	Binary Signed Short Price	The limit order price
<i>Reserved</i>	25	1	Reserved	Reserved
Total Length = 26 bytes				

Table 13. Add Order (expanded)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x2F	Add Order Message (expanded)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	Day-specific identifier assigned to this order
<i>Side Indicator</i>	14	1	Alphanumeric	B = Buy Order

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
				S = Sell Order
<i>Quantity</i>	15	4	Binary	Instrument quantity being added to the complex book (may be less than the number entered).
<i>Complex Instrument Id</i>	19	8	Printable ASCII	Complex Instrument Id right padded with spaces.
<i>Price</i>	27	8	Binary Signed Long Price	The limit order price
<i>Reserved</i>	35	1	Reserved	Reserved
<i>Participant ID</i>	36	4	Alphanumeric	<i>Optionally specified.</i> If specified the Executing Broker of firm attributed to this quote. Space filled otherwise.
<i>Customer Indicator</i>	40	1	Alphanumeric	N = Non-Customer C = Customer
<i>Client ID</i>	41	4	Alphanumeric	Optional user specified value attributed to this quote. Space filled otherwise.
<i>Reserved</i>	45	5	Reserved	Reserved
Total Length = 50 bytes				

Order Modification Messages Message Fields

Order Modification messages refer to an Order ID previously sent with an **Add Order** message. Multiple Order Modification messages may modify a single complex order and the effects are cumulative. Modify messages may update the size and/or the price of a complex order on the book. When the remaining size of a complex order reaches zero, the complex order is dead and should be removed from the book.

Order Executed

Order Executed messages are sent when a visible complex order on the Cboe complex book is executed in whole or in part. The execution price equals the limit order price found in the original **Add Order** message or the limit order price in the latest **Modify Order** message referencing the *Order Id*.

Note even if there are single leg to complex order executions, this feed will only contain the order execution for the complex order. Any single leg execution information is available on the standard PITCH feed.

Table 14. Order Executed

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x23	Order Executed Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of a previously sent Add Order message that was executed
<i>Executed Quantity</i>	14	4	Binary	Instrument quantity executed
<i>Execution Id</i>	18	8	Binary	Cboe generated day-unique execution identifier of this execution.
<i>Trade Condition</i>	26	1	Alphanumeric	See Options Trade Condition Codes on page 121 for details about new codes.
Total Length = 27 bytes				

Order Executed at Price/Size

Order Execution at Price/Size messages are sent when a complex order on the Cboe complex book is executed in whole or in part at a different price than the limit price on the original **Add Order** message or the limit order price in the latest **Modify Order** message referencing the *Order Id*. If the *Remaining Quantity* field contains a 0 the complex order should be completely removed from the complex book.

Order Execution at Price/Size messages may also be sent in the event the existing size for Order Id is not equal to *Executed Quantity* + *Remaining Quantity*. In this case the complex order should be prioritized the same as a new complex order.

Table 15. Order Executed at Price/Size

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x24	Order Executed at Price/Size Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of a previously sent Add Order message that was executed
<i>Executed Quantity</i>	14	4	Binary	Instrument quantity executed
<i>Remaining Quantity</i>	18	4	Binary	Number of contracts remaining after the execution
<i>Execution Id</i>	22	8	Binary	Cboe generated day-unique execution identifier of this execution.
<i>Price</i>	30	8	Binary Signed Long Price	The execution price of the order
<i>Trade Condition</i>	38	1	Alphanumeric	See Options Trade Condition Codes on page 121 for details about new codes.
Total Length = 39 bytes				

Reduce Size

Reduce Size messages are sent when a complex order on the Cboe complex book is partially reduced.

Table 16. Reduce Size (long)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x25	Reduce Size Message (long)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of a previously sent Add Order message that has been reduced
<i>Canceled Quantity</i>	14	4	Binary	Instrument quantity canceled
Total Length = 18 bytes				

Table 17. Reduce Size (short)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x26	Reduce Size Message (short)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of a previously sent Add Order message that has been reduced
<i>Canceled Quantity</i>	14	2	Binary	Instrument quantity canceled
Total Length = 16 bytes				

Modify Order

The **Modify Order** message is sent whenever an open complex order is visibly modified. The **Order Id** refers to the **Order Id** of the original **Add Order** message.

Note that **Modify Order** messages that appear to be "no ops" (i.e. they do not appear to modify any relevant fields) will still lose priority.

Table 18. Modify (long)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x27	Modify Order message (long)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of a previously sent Add Order message that has been modified
<i>Quantity</i>	14	4	Binary	Instrument quantity associated with this complex order after this modify (may be less than the number entered)
<i>Price</i>	18	8	Binary Signed Long Price	The limit order price after this modify
<i>Reserved</i>	26	1	Bit Field	Reserved
Total Length = 27 bytes				

Table 19. Modify (short)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x28	Modify Order message (short)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of a previously sent Add Order message that has been modified
<i>Quantity</i>	14	2	Binary	Instrument quantity associated with this complex order after this modify (may be less than the number entered)
<i>Price</i>	16	2	Binary Signed Short Price	The limit order price after this modify
<i>Reserved</i>	18	1	Bit Field	Reserved
Total Length = 19 bytes				

Delete Order

The **Delete Order** message is sent whenever a booked order is cancelled or leaves the order book. The *Order Id* refers to the *Order Id* of the original **Add Order** message. An order that is deleted from the book may return to the book later under certain circumstances. Therefore, a **Delete Order** message does not indicate that a given *Order Id* will not be sent again on a subsequent **Add Order** message.

Table 20. Delete

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x29	Delete Order Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of a previously sent Add Order message that has been removed from order book.
Total Length = 14 bytes				

Trade Message Fields

The **Trade** message provides information about executions of complex order auctions on the Cboe complex book. A **Trade** message can also be sent when an auction executes against a non-displayed order, such as a contra response. **Trade** messages are necessary to calculate Cboe execution-based data. **Trade** messages do not alter the complex book and can be ignored if messages are being used solely to build a complex book.

No **Add Order** message is sent for complex auction orders, and thus, no order modification messages may be sent when complex auctions are executed. Instead, a **Trade** message is sent whenever a complex auction is executed in whole or in part. A complete view of all Cboe complex executions can be built by combining all **Order Executed** messages and **Trade** messages.

Table 21. Trade (long)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x2A	Trade Message (long)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of the executed order.
<i>Side Indicator</i>	14	1	Alphanumeric	Always B=Buy Order regardless of resting side
<i>Quantity</i>	15	4	Binary	Instrument quantity traded
<i>Complex Instrument Id</i>	19	6	Printable ASCII	<i>Complex Instrument Id</i> right padded with spaces.
<i>Price</i>	25	8	Binary Signed Long Price	The execution price of the order
<i>Execution Id</i>	33	8	Binary	Cboe generated day-unique execution identifier of this trade.
<i>Trade Condition</i>	41	1	Alphanumeric	See Options Trade Condition Codes on page 121 for details about new codes.
Total Length = 42 bytes				

Table 22. Trade (short)

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x2B	Trade Message (short)
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Order Id</i>	6	8	Binary	<i>Order Id</i> of the executed order.
<i>Side Indicator</i>	14	1	Alphanumeric	Always B=Buy Order regardless of resting side
<i>Quantity</i>	15	2	Binary	Instrument quantity traded
<i>Complex Instrument Id</i>	17	6	Printable ASCII	<i>Complex Instrument Id</i> right padded with spaces.
<i>Price</i>	23	2	Binary Signed Short Price	The execution price of the order
<i>Execution Id</i>	25	8	Binary	Cboe generated day-unique execution identifier of this trade.

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
Trade Condition	33	1	Alphanumeric	See Options Trade Condition Codes on page 121 for details about new codes.
Total Length = 34 bytes				

Auction Notification Message Fields

Auction Notification messages are used to disseminate order details of a complex auction. Auctions will be available for a defined period of time known as the exposure period.

Table 23. Auction Notification

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	Length of this message including this field
<i>Message Type</i>	1	1	0xAD	Auction Notification message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp.
<i>Complex Instrument Id</i>	6	6	Printable ASCII	Complex Instrument Id right padded with spaces.
<i>Auction ID</i>	12	8	Binary	Day specific identifier assigned to this auction.
<i>Auction Type</i>	20	1	Alphanumeric	C = Complex Order Auction (COA) (or Stop Complex Order Auction (SCOA) effective TBD) S = Complex Solicitation Auction Mechanism B = Complex AIM O = COA All or None
<i>Side</i>	21	1	Alphanumeric	B = Buy S = Sell
<i>Price</i>	22	8	Binary Signed Long Price	Auction price. The price field will be populated for all Auctions on EDGX Options, and for SAM Auctions on C1. This field will reflect the auction start price for SPX and SPXW AIM (C1 Only) and will be set to zero for all other AIM on C1. This field will be set to zero for COA on C1, C2, and BZX Options.
<i>Quantity</i>	30	4	Binary	Instrument quantity.
<i>Customer Indicator</i>	34	1	Alphanumeric	N = Non-Customer C = Customer
<i>ParticipantID</i>	35	4	Alphanumeric	Executing Broker (optional) of firm attributed to this quote.
<i>Auction End Offset</i>	39	4	Binary	Nanosecond offset from last timestamp.
<i>Client ID</i>	43	4	Alphanumeric	Optional user specified value attributed to this quote. Space filled otherwise.
Total Length = 47 bytes				

Auction Cancel Message Fields

Auction Cancel messages are used to disseminate the cancelation of an earlier **Auction Notification** message as a result of a user cancelation of the original complex auction, a user modification request to change the complex auction price or increase the original complex auction quantity, a fading of the NBBO or to cancel any remaining complex auction quantity from the original **Auction Notification** following the complex auction termination.

A user request to modify the complex auction price or to increase the original complex auction quantity will result in a cancelation of the complex auction followed by a new **Auction Notification** message. **Auction Cancel** messages will not be issued for complex auction quantity decrements.

Table 24. Auction Cancel

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	Length of this message including this field
<i>Message Type</i>	1	1	0xAE	Auction Cancel message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Auction ID</i>	6	8	Binary	Day specific identifier assigned to this auction
Total Length = 14 bytes				

Auction Trade Message Fields

Auction Trade messages are used to disseminate executions resulting from a complex auction.

Table 25. Auction Trade

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	Length of this message including this field
<i>Message Type</i>	1	1	0xAF	Auction Trade message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp.
<i>Auction ID</i>	6	8	Binary	Day specific identifier assigned to this auction
<i>Execution ID</i>	14	8	Binary	Day specific identifier assigned to this execution
<i>Price</i>	22	8	Binary Signed Long Price	Trade price
<i>Quantity</i>	30	4	Binary	Instrument quantity traded
Total Length = 34 bytes				

Trading Status Message Fields

The **Trading Status** message is used to indicate the current trading status of a complex instrument. A **Trading Status** message will be sent whenever a complex instrument trading status changes.

A **Trading Status** message will be sent for all complex instruments as they transition through various trading states.

Starting at 7:30 a.m. ET, Cboe will send a *Trading Status* of Q once orders can be accepted for queuing in preparation for the RTH open. At or after 9:30 a.m. ET, Cboe will send a *Trading Status* of T as series are open for trading. Cboe will send a *Trading Status* of L as SPX or VIX series transition from RTH trading to Curb trading.

A **Trading Status** message will also be sent:

- for a Regulatory Halt Q=Quotingperiod in any series where the underlying has experienced a Regulatory Halt as well as the T=Trading resumption for the same series.
- for instruments that are in a Q=Quoting period for auctions.

The *Trading Status* field will be used to represent the status of the RTH (9:30 a.m. ET - 4:15 p.m. ET) and Curb sessions. The *GTH Trading Status* field will be used to represent the status of series that trade during the GTH session (C1 Only). The GTH session runs from 8:15 p.m. to 9:25 a.m. ET for SPX and VIX series (C1 only).

Table 26. Trading Status

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	Length of this message including this field
<i>Message Type</i>	1	1	0x31	Trading Status message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp
<i>Complex Symbol ID</i>	6	6	Printable ASCII	<i>Complex Symbol</i> right padded with spaces.
<i>Reserved</i>	12	2	Reserved	Reserved
<i>Trading Status</i>	14	1	Alpha	H = Halted L = Curb Trading (C1 Only) Q = Quote-Only T = RTH Trading
<i>Reserved</i>	15	1	Reserved	Reserved
<i>GTH Trading Status</i> (C1 Only)	16	1	Alpha	H = Halted Q = Quote-Only T = Trading
<i>Reserved2</i>	17	1	Alpha	Reserved
Total Length = 18 bytes				

Options Auction Update Message Fields

Options Auction Update messages are used to disseminate price and size information during the Opening and Re-Opening (halt) process for complex instruments. The **Options Auction Update** messages are sent every 5 seconds during an opening period. Refer to the [Cboe Options Complex Book Process](#) specification for more information.

Table 27. Options Auction Update

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field.
<i>Message Type</i>	1	1	0xD1	Options Auction Update Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp.
<i>Complex Instrument ID</i>	6	8	Printable ASCII	<i>Complex Instrument</i> right padded with spaces.
<i>Auction Type</i>	14	1	Alphanumeric	G = GTH Opening (C1 Only) O = RTH Opening (C1 Only) H = Halt Re-Opening
<i>Reference Price</i>	15	8	Binary Long Price	Not used for complex series. Will contain zero value.
<i>Buy Contracts</i>	23	4	Binary	Cumulative Buy interest at the Indicative Price.
<i>Sell Contracts</i>	27	4	Binary	Cumulative Sell interest at the Indicative Price.
<i>Indicative Price</i>	31	8	Binary Signed Long Price	SNBBO Collared Volume Maximizing Imbalance Minimizing Price computed on combined Auction-Only and Continuous Book (if any).
<i>Auction Only Price</i>	39	8	Binary Signed Long Price	Not used for complex series. Will contain zero value.
<i>Opening Condition</i>	47	1	Alphanumeric	Not used for Complex series. Will contain zero value.
<i>Composite Market Bid Price</i>	48	8	Binary Signed Long Price	Not used for Complex series. Will contain zero value.
<i>Composite Market Offer Price</i>	56	8	Binary Signed Long Price	Not used for complex series. Will contain zero value.
Total Length = 64 bytes				

Auction Summary Message Fields

Auction Summary messages are used to disseminate the results of an auction of a complex instrument. An Opening or Re-Opening **Auction Summary** message for each complex instrument is sent at the conclusion of its Opening or Re-Opening auction and represents Cboe opening price. Refer to the [Cboe Options Complex Book Process](#) specification for more information.

The **Auction Summary** message has the following format:

Table 28. Auction Summary

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	Length of this message including this field.
<i>Message Type</i>	1	1	0x96	Auction Summary Message
<i>Time Offset</i>	2	4	Time Offset	Nanosecond offset from last unit timestamp.
<i>Complex Instrument Id</i>	6	8	Printable ASCII	<i>Complex Instrument Id</i> right padded with spaces.
<i>Auction Type</i>	14	1	Alphanumeric	G = GTH Opening (C1 Only) O = RTH Opening (C1 Only) H = Halt Re-Opening
<i>Price</i>	15	8	Binary Signed Long Price	Auction price
<i>Quantity</i>	23	4	Binary	Cumulative instrument quantity executed during the auction
Total Length = 27 bytes				

End of Session Message Fields

The **End of Session** message is sent for each unit when the unit shuts down. No more sequenced messages will be delivered for this unit, but heartbeats from the unit may be received.

Table 29. End of Session

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x2D	End of Session message
<i>Timestamp</i>	2	4	Binary	Nanosecond offset from last unit timestamp
Total Length = 6 bytes				

Gap Request Proxy Messages

The following messages are used for initializing a TCP/IP connection to the Gap Request Proxy (GRP) and to request message retransmissions. Customers only need to implement the following messages if gap requests will be made. The following messages will not be delivered using multicast.

Login Message Fields

The **Login** message is the first message sent to the GRP by a user's process after the connection to the GRP is established. Failure to login before sending any other message type will result in the connection being dropped by the GRP.

Table 30. Login

FIELD	OFFSET	LENGTH	VALUE/TYPE	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x01	Login message
<i>SessionSubId</i>	2	4	Alphanumeric	<i>SessionSubId</i> supplied by Cboe
<i>Username</i>	6	4	Alphanumeric	<i>Username</i> supplied by Cboe
<i>Filler</i>	10	2	Alphanumeric	(space filled)
<i>Password</i>	12	10	Alphanumeric	<i>Password</i> supplied by Cboe
Total Length = 22 bytes				

Login Response Message Fields

The **Login Response** message is sent by the GRP to a user’s process in response to a **Login** message. The status field is used to reflect an accepted login or the reason the session was not accepted. If login fails, the connection will be dropped after the **Login Response** message is sent.

Table 31. Login Response

FIELD	OFFSET	LENGTH	VALUE/TYPE	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x02	Login Response Message
<i>Status</i>	2	1	Alphanumeric	Accepted or reason for reject
Total Length = 3 bytes				
Login Response - Status Codes				
'A'	Login Accepted			
'N'	Not authorized (Invalid Username/Password)			
'B'	Session in use			
'S'	Invalid Session			

Gap Request Message Fields

The **Gap Request** message is used by a user's process to request retransmission of a sequenced message (or messages) by one of Cboe's gap servers.

Table 32. Gap Request

FIELD	OFFSET	LENGTH	VALUE/TYPE	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x03	Gap Request message
<i>Unit</i>	2	1	Binary	<i>Unit</i> that the gap is requested for
<i>Sequence</i>	3	4	Binary	<i>Sequence</i> of first message (lowest sequence in range)
<i>Count</i>	7	2	Binary	<i>Count</i> of messages requested
Total Length = 9 bytes				

Gap Response Message Fields

The **Gap Response** message is sent by the GRP in response to a **Gap Request** message. The *Unit* and *Sequence* fields will match the values supplied in the **Gap Request** message. A **Gap Response** message, with a Status of Accepted or reason for failure, will be sent for each **Gap Request** message received by the GRP.

Table 33. Gap Response

FIELD	OFFSET	LENGTH	VALUE/TYPE	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x04	Gap Response message
<i>Unit</i>	2	1	Binary	<i>Unit</i> the gap was requested for
<i>Sequence</i>	3	4	Binary	<i>Sequence</i> of first message in request
<i>Count</i>	7	2	Binary	<i>Count</i> of messages requested
<i>Status</i>	9	1	Alphanumeric	Accepted or reason for reject
Total Length = 10 bytes				
Gap Response - Status Codes				
'A'	Accepted			
'O'	Out of range (ahead of sequence or too far behind)			
'D'	Daily gap request allocation exhausted			
'M'	Minute gap request allocation exhausted			
'S'	Second gap request allocation exhausted			
'C'	Count request limit for one gap request exceeded			
'I'	Invalid Unit specified in request			
'U'	Unit is currently unavailable			

* All non-'A' status codes should be interpreted as a reject.

Spin Messages

Login Message Fields

The **Login** message is the first message sent to the Spin Server by a user's process after the connection to the Spin Server is established. Failure to login before sending any other message type will result in the connection being dropped by the Spin Server.

The format of the **Login** message for the Spin Server is identical to that of the GRP described in [Login Message Fields](#) on page 43.

Login Response Message Fields

The **Login Response** message is sent by the Spin Server to a user's process in response to a **Login** message. The status field is used to reflect an accepted login or the reason the session was not accepted. If login fails, the connection will be dropped after the **Login Response** message is sent.

The format of the **Login Response** message for the Spin Server is identical to that of the GRP described in [Login Response Message Fields](#) on page 44.

Spin Image Available Message Fields

The **Spin Image Available** message is sent once per second and indicates through what sequence number a spin is available.

Table 34. Spin Image Available

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x80	Spin Image Available message
<i>Sequence</i>	2	4	Binary	Spin is available which is current through this sequence number
Total Length = 6 bytes				

Spin Request Message Fields

The **Spin Request** message is used by a user's process to request transmission of a spin of the unit's order book. Refer to [Spin Servers](#) on page 12 for more complete details regarding *Sequence* specification as well as buffering requirements.

Table 35. Spin Request

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x81	Spin Request message
<i>Sequence</i>	2	4	Binary	Sequence number from a Spin Image Available message received by the customer
Total Length = 6 bytes				

Spin Response Message Fields

The **Spin Response** message is sent in response to a user's **Spin Request** message indicating whether a spin will be sent.

Table 36. Spin Response

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x82	Spin Response message
<i>Sequence</i>	2	4	Binary	Sequence number from a Spin Image Available message received by the customer
<i>Order Count</i>	6	4	Binary	Number of Add Order messages which will be contained in this spin
<i>Status</i>	10	1	Alphanumeric	Accepted or reason for reject
Total Length = 11 bytes				
Spin Response - Status Codes				
'A'	Accepted			
'O'	Out of Range (<i>Sequence</i> requested is greater than <i>Sequence</i> available by the next spin)			
'S'	Spin already in progress (only one spin can be running at a time)			

* All non-'A' status codes should be interpreted as a reject.

Spin Finished Message Fields

The **Spin Finished** message is sent to indicate that all messages for the spin requested have been sent. A **Spin Finished** message is only sent if a **Spin Request** was not rejected. Upon receipt of a **Spin Finished** message, any buffered multicast messages should be applied to the customer’s copy of the book to make it current.

Table 37. Spin Finished

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x83	Spin Finished message
<i>Sequence</i>	2	4	Binary	Sequence number from the Spin Request message
Total Length = 6 bytes				

Instrument Definition Request Message Fields

The **Instrument Definition Request** message is used by a user's process to request transmission of this unit's Symbol Mappings and Complex Instrument Definitions. All **Symbol Mapping** messages will be sent before **Complex Instrument Definition Expanded** and **Exchange Designated Complex Instrument Definition** messages. Refer to Section 1.6 for more complete details regarding *Sequence* specification as well as buffering requirements.

Table 38. Instrument Definition Request

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x84	Instrument Definition Request message
<i>Sequence</i>	2	4	Binary	Must be 0. Only the current Symbol Mappings and Complex Instrument Definitions are available.
Total Length = 6 bytes				

Instrument Definition Response Message Fields

The **Instrument Definition Response** message is sent in response to a user's **Instrument Definition Request** message indicating whether a spin will be sent.

Table 39. Instrument Definition Response

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	Length of this message including this field
<i>Message Type</i>	1	1	0x85	Instrument Definition Response message
<i>Sequence</i>	2	4	Binary	Will always be 0.
<i>Instrument Count</i>	6	4	Binary	Number of Symbol Mapping and Complex Instrument Definition (if applicable) messages which will be contained in this spin
<i>Status</i>	10	1	Alphanumeric	Accepted or reason for reject
Total Length = 11 bytes				
Instrument Definition Response - Status Codes				
'A'	Accepted			
'O'	Out of Range (<i>Sequence</i> must be 0)			
'S'	Spin already in progress (only one spin can be running at a time)			

* All non-'A' status codes should be interpreted as a reject.

Instrument Definition Finished Message Fields

The **Instrument Definition Finished** message is sent to indicate that all **Symbol Mapping** and **Complex Instrument Definition Expanded** messages for this unit have been sent. An **Instrument Definition Finished** message is only sent if an **Instrument Definition Request** was not rejected.

Table 40. Instrument Definition Finished

FIELD NAME	OFFSET	LENGTH	TYPE/(VALUE)	DESCRIPTION
<i>Length</i>	0	1	Binary	<i>Length</i> of this message including this field
<i>Message Type</i>	1	1	0x86	Instrument Definition Finished Message
Total Length = 2 bytes				

Spin Server Usage Example

The following diagram shows the exchange of messages over time between a customer and Cboe's Multicast PITCH feed and spin server. Note that while the example may seem to imply only **Complex Instrument Definition Expanded**, **Time** and **Add Order** messages would be sent on a spin, this is not the case. **Trading Status** and **Auction Update** messages may also be sent.

At time 1, the customer has no state of the book and desires to become current. The customer caches the received Multicast PITCH messages (sequences 310172 and 310173) for later use. Since the customer has no book, they cannot yet be applied.

At time 5, the customer has successfully logged into the Spin Server and has cached another message, sequence 310174.

At time 7, the customer receives a **Spin Image Available** message which indicates that the spin server is capable of giving them a spin of all open orders as of sequence 310169. The customer does not have all messages cached after 310169 (they are missing 310170 and 310171), so this spin is not useful to the customer.

At time 10, the customer receives a **Spin Image Available** message which is useful since it would be a spin of all orders up to and including sequence 310175 and the customer has all messages after 310175 cached.

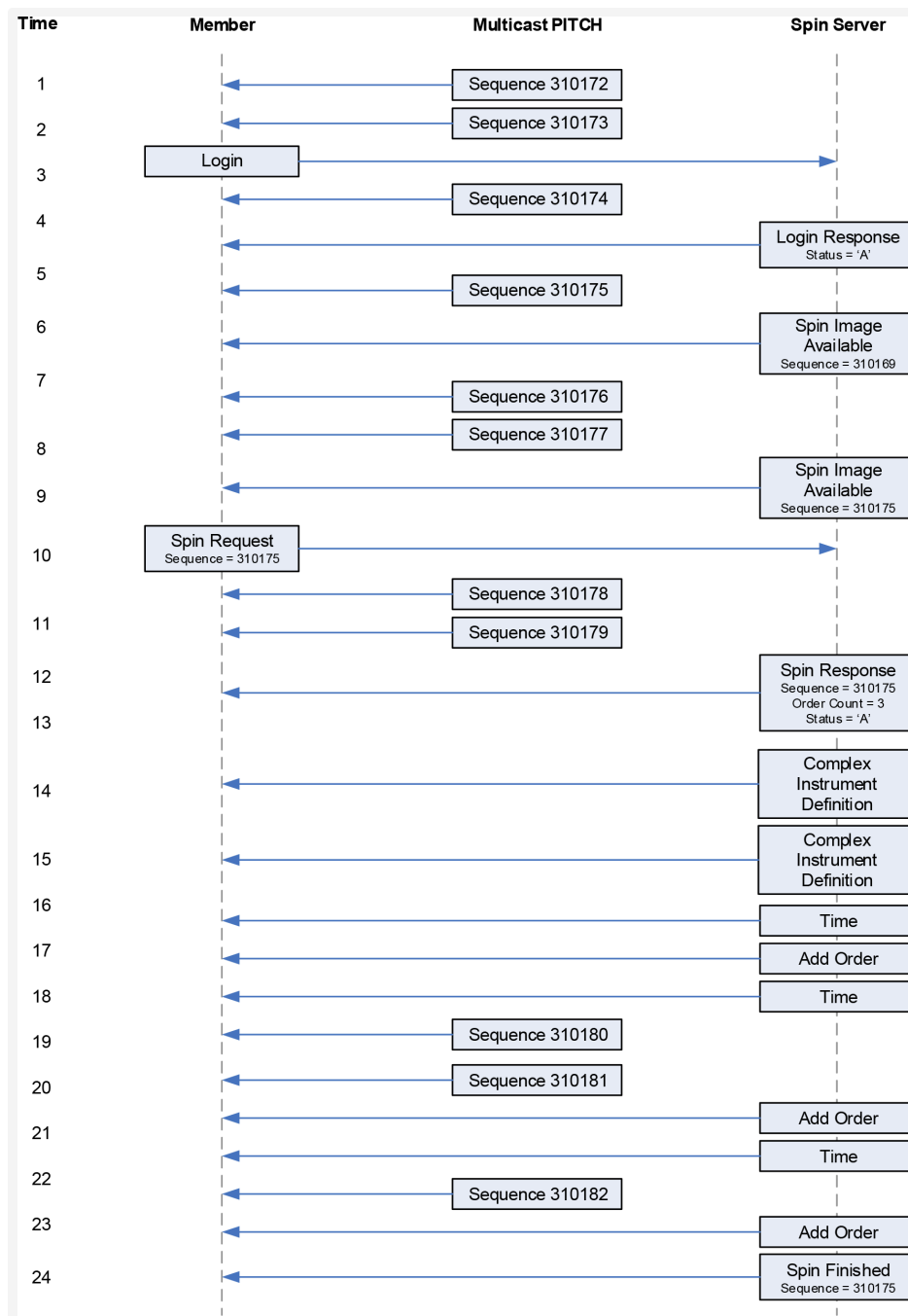
At time 11, the customer sends a **Spin Request** for all messages up to and including 310175 and continues to cache Multicast PITCH messages received.

At time 14, the spin server acknowledges the spin request and indicates that five messages will be sent (complex instrument definition and open orders).

At time 24, the spin server indicates that it has finished sending all messages. The customer must then apply the cached messages from sequence number 310176 through current.

Notes:

- Spin Servers are available for each unit. Customers may need to employ multiple Spin Servers depending upon their architecture.
- As a rule of thumb, in its Options markets, Cboe typically has ~300,000 complex instruments defined and ~3.2 million open orders across all units, or an average of about 9,375 complex instruments and 100,000 open orders per unit. The actual number per unit varies depending upon activity in individual symbols. Expect this number to increase and plan accordingly.



Message Types

Gap Request Proxy Messages

0x01	Login
0x02	Login Response
0x03	Gap Request
0x04	Gap Response

Spin Server Messages

0x01	Login
0x02	Login Response
0x80	Spin Image Available
0x81	Spin Request
0x82	Spin Response
0x83	Spin Finished
0x84	Instrument Definition Request
0x85	Instrument Definition Response
0x86	Instrument Definition Finished

PITCH 2.X Messages

0xB1	Time Reference (C1 Only)
0x20	Time
0x97	Unit Clear
0xBC	Transaction Begin
0xBD	Transaction End
0x9A	Complex Instrument Definition Expanded
0x9F	Exchange Designated Complex Instrument Definition (C1 Only)
0x2F	Symbol Mapping
0x21	Add Order - Long
0x22	Add Order - Short
0x2F	Add Order - Expanded
0x23	Order Executed
0x24	Order Executed at Price/Size
0x25	Reduce Size - Long
0x26	Reduce Size - Short
0x27	Modify Order - Long
0x28	Modify Order - Short
0x29	Delete Order
0x2A	Trade - Long
0x2B	Trade - Short
0xAD	Auction Notification
0xAE	Auction Cancel
0xAF	Auction Trade
0x31	Trading Status
0xD1	Options Auction Update
0x96	Auction Summary
0x2D	End of Session

Example Messages

Each of the following message types must be wrapped by a sequenced or unsequenced unit header as described in [Sequenced Unit Header](#) on page 17. Note that in the following examples, each byte is represented by two hexadecimal digits.

Login Message Example

Table 41. Login Message Example

Length	16	22 bytes
Type	01	Login
SessionSubId	30 30 30 31	"0001"
Username	46 49 52 4D	"FIRM"
Filler	20 20	" "
Password	41 42 43 44 30 30 20 20 20 20	"ABCD00"

Login Response Message Example

Table 42. Login Response Message Example

Length	03	3 bytes
Type	02	Login Response
Status	41	Login accepted

Gap Request Message Example

Table 43. Gap Request Message Example

Length	09	9 bytes
Type	03	Gap Request
Unit	01	Unit 1
Sequence	3B 10 00 00	First message: 4155
Count	32 00	50 messages

Gap Response Message Example

Table 44. Gap Response Message Example

Length	10	10 bytes
Type	04	Gap Response
Unit	01	Unit 1
Sequence	3B 10 00 00	First message: 4155
Count	32 00	50 messages
Status	41	Accepted

Spin Image Available Message Example

Table 45. Spin Image Available Message Example

Length	06	6 bytes
Type	80	Spin Image Available
Sequence	3B 10 00 00	Sequence: 4155

Spin Request Message Example

Table 46. Spin Request Message Example

Length	06	6 bytes
Type	81	Spin Request
Sequence	3B 10 00 00	Sequence: 4155

Spin Response Message Example

Table 47. Spin Response Message Example

Length	0B	11 bytes
Type	82	Spin Request
Sequence	3B 10 00 00	Sequence: 4155
Order Count	42 00 00 00	66 orders
Status	41	Accepted

Spin Finished Message Example

Table 48. Spin Finished Message Example

Length	06	6 bytes
Type	83	Spin Finished
Sequence	3B 10 00 00	Sequence: 4155

Instrument Definition Request Message Example

Table 49. Instrument Definition Request Message Example

Length	06	6 bytes
Type	84	Instrument Definition Request
Sequence	00 00 00 00	Sequence: 0

Instrument Definition Response Message Example

Length	0B	11 bytes
Type	85	Instrument Definition Response
Sequence	00 00 00 00	Sequence: 0
Instrument Count	B8 0B 00 00	3000 Instruments
Status	41	Accepted

Instrument Definition Finished Message Example

Table 50. Instrument Definition Finished Message Example

Length	02	2 bytes
Type	86	Instrument Definition Finished

Time Reference Message Example (C1 Only)

Table 51. Time Reference Message Example (C1 Only)

Length	12	18 bytes
Type	B1	Time Reference
Midnight	D0 8B 34 60	2021-02-23 00:00:00
Reference		Eastern (1614056400 seconds since the Epoch)
Time	00 E1 00 00	16:00:00
Time Offset	00 00 00 00	Exactly 16:00:00
Trade Date	2F 62 34 01	2021-02-23 February 23, 2021

Time Message Example

Table 52. Time Message Example

Length	06	6 bytes
Type	20	Time
Time	98 85 00 00	34,200 seconds = 09:30 AM Eastern

Time Message Example

Table 53. Time Message Example

Length	10	0A bytes
Type	20	Time
Time	98 85 00 00	34,200 seconds =09:30 AM Eastern
Epoch Time (C1 Only)	68 11 35 60	1,614,090,600 seconds since the Epoch

Unit Clear Message Example

Table 54. Unit Clear Message Example

Length	06	6 bytes
Type	97	Unit Clear
Time Offset	18 D2 06 00	447,000 ns since last Time Message

Add Order - Long Message Example

Table 55. Add Order - Long Message Example

Length	22	34 bytes
Type	21	Add Order - Long
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Side Indicator	42	Buy
Quantity	32 00 00 00	50
CID	43 30 30 30 31 32	C00012
Price	28 23 00 00 00 00 00 00	\$0.9000
Reserved	00	Reserved

Add Order - Short Message Example

Table 56. Add Order - Short Message Example

Length	1A	26 bytes
Type	22	Add Order - Short
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Side Indicator	42	Buy
Quantity	32 00	50
CID	43 30 30 30 31 32	C00012
Price	0A 28	\$102.50
Reserved	00	Reserved

Add Order - Expanded Message Example

Table 57. Add Order - Expanded Message Example

Length	2D	45 bytes
Type	2F	Add Order - Expanded
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Side Indicator	42	Buy
Quantity	32 00	50
CID	43 30 30 30 31 32 20 20	C00012
Price	28 23 00 00 00 00 00 00	\$0.9000
Reserved	00	Reserved
Participant ID	41 42 43 44	ABCD
Customer Indicator	4E	Non-Customer
Client ID	43 4C 49 44	CLID

Order Executed Message Example

Table 58. Order Executed Message Example

Length	1A	26 bytes
Type	23	Order Executed
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Executed Quantity	64 00 00 00	100
Execution Id	34 2B 46 E0 BB 00 00 00	0AAP09VEC

Order Executed at Price/Size Message Example

Table 59. Order Executed at Price/Size Message Example

Length	26	38 bytes
Type	24	Order Executed at Price/Size
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Executed Quantity	64 00 00 00	100
Remaining	32 00 00 00	50
Execution Id	34 2B 46 E0 BB 00 00 00	0AAP09VEC
Price	E8 A3 0F 00 00 00 00 00	\$102.50

Reduce Size - Long Message Example

Table 60. Reduce Size - Long Message Example

Length	12	18 bytes
Type	25	Reduce Size - Long
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Canceled Quantity	64 00 00 00	100

Reduce Size - Short Message Example

Table 61. Reduce Size - Short Message Example

Length	10	16 bytes
Type	26	Reduce Size - Short
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Canceled Quantity	64 00	100

Modify Order - Long Message Example

Table 62. Modify Order - Long Message Example

Length	1B	27 bytes
Type	27	Modify Order - Long
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Quantity	4B 00 00 00	75
Price	E8 A3 0F 00 00 00 00 00	\$102.50
Reserved	00	Reserved

Modify Order - Short Message Example

Table 63. Modify Order - Short Message Example

Length	13	19 bytes
Type	28	Modify Order - Short
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Quantity	4B 00	75
Price	0A 28	\$102.50
Reserved	00	Reserved

Delete Order Message Example

Table 64. Delete Order Message Example

Length	0E	14 bytes
Type	29	Delete Order
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	

Trade - Long Message Example

Table 65. Trade - Long Message Example

Length	29	41 bytes
Type	2A	Trade - Long
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Side	42	Buy
Quantity	4B 00 00 00	75
CID	43 30 30 30 31 32	C00012
Price	E8 A3 0F 00 00 00 00 00	\$102.50
Execution Id	34 2B 46 E0 BB 00 00 00	0AAP09VEC

Trade - Short Message Example

Table 66. Trade - Short Message Example

Length	21	33 bytes
Type	2B	Trade - Short
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Side	42	Buy
Quantity	64 00	100
CID	43 30 30 30 31 32	C00012
Price	0A 28	\$102.50
Execution Id	34 2B 46 E0 BB 00 00 00	0AAP09VEC

Auction Notification Message Example

Table 67. Auction Notification Message Example

Length	2F	47 bytes
Type	AD	Auction Notification
Time Offset	18 D2 06 00	447,000 ns since last Time Message
CID	43 30 30 30 31 32	C00012
Auction ID	05 40 5B 77 8F 56 1D 0B	631WC4000005
Auction Type	4F	O = COA AON
Side	42	B = Buy Side
Price	00 00 00 00 00 00 00 00	Price not displayed
Quantity	64 00 00 00	100
Customer Indicator	43	C = Customer
Participant ID	45 46 49 44	EFID
Auct. End Offset	38 73 0E 00	947,000 ns since last Time Message
Client ID	43 4C 49 44	CLID

Auction Cancel Message Example

Table 68. Auction Cancel Message Example

Length	E	14 bytes
Type	AE	Auction Cancel
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Auction ID	05 40 5B 77 8F 56 1D 0B	631WC4000005

Auction Trade Message Example

Table 69. Auction Trade Message Example

Length	22	34 bytes
Type	AF	Auction Trade
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Auction ID	05 40 5B 77 8F 56 1D 0B	631WC4000005
Execution Id	34 2B 46 E0 BB 00 00 00	0AAP09VEC
Price	E8 A3 0F 00 00 00 00 00	\$102.50
Quantity	64 00 00 00	100

End of Session Message Example

Table 70. End of Session Message Example

Length	06	6 bytes
Type	2D	End of Session
Time Offset	18 D2 06 00	447,000 ns since last Time Message

Trading Status Message Example

Table 71. Trading Status Message Example

Length	12	18 bytes
Type	31	Trading Status
Time Offset	18 D2 06 00	447,000 ns since last Time Message
CID	39 39 38 38 37 37	998877
Reserved	20 20	Reserved
Trading Status	54	T = Trading
Reserved	20	Reserved
Global Trading	48	H = Halted
Hours Status	Reserved	20
Reserved	20	Reserved

Sequenced Unit Header with 2 Messages Example

Table 72. Sequenced Unit Header

Hdr Length	31 00	49 bytes, including header
Hdr Count	02	2 messages to follow
Hdr Unit	01	Unit 1
Hdr Sequence	01 00 00 00	First message has sequence number 1

Table 73. Message 1: Add Order (Short)

Length	1A	26 bytes
Message format	22	Add Order - Short
Time Offset	18 D2 06 00	447,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Side Indicator	42	Buy
Quantity	E1 02	737
CID	43 30 30 30 31 32	C00012
Price	01 00	0.01
Reserved	00	Reserved

Table 74. Message 2: Reduce Size (Short)

Length	10	16 bytes
Message format	26	Reduce Size - Short
Time Offset	E8 D9 06 00	449,000 ns since last Time Message
Order Id	05 40 5B 77 8F 56 1D 0B	631WC4000005
Canceled Quantity	E1 02	737

Options Auction Update Message Example

Table 75. Options Auction Update Message Example

Length	40	64 bytes
Type	D1	Options Auction Update
Time Offset	18 D2 06 00	447,000 ns since last Time Message
CID	43 30 30 30 31 32 20 20	C00012
Auction Type	4F	RTH Opening
Reference Price	00 00 00 00 00 00 00 00	always zero
Buy Contracts	64 00 00 00	100 Contracts
Sell Contracts	C8 00 00 00	200 Contracts
Indicative Price	E8 A3 0F 00 00 00 00 00	\$102.50
Auction Only Price	00 00 00 00 00 00 00 00	always zero
Opening Condition	00	always zero
Composite Market Bid Price	00 00 00 00 00 00 00 00	always zero
Composite Market Offer Price	00 00 00 00 00 00 00 00	always zero

Auction Summary Message Example

Table 76. Auction Summary Message Example

Length	1B	27 bytes
Type	96	Auction Summary
Time Offset	18 D2 06 00	447,000 ns since last Time Message
CID	43 30 30 30 31 32 20 20	C00012
Auction Type	4F	RTH Opening
Price	E8 A3 0F 00 00 00 00 00	\$102.50
Quantity	4B 00 00 00	75

Complex Instrument Definition Expanded Message Example

Table 77. Complex Instrument Definition Expanded Message Example

Length	33	51 bytes
Type	9A	Complex Instrument Definition Expanded
Time Offset	18 D2 06 00	447,000 ns since last Time Message
CID	43 30 30 30 31 32	C00012
Complex Instrument	5A 56 5A 5A 54 20 20	ZVZZT
Underlying Complex Instrument	4F 00 00 00	0 = All Legs are Options
Type	Leg Count	02
Leg Symbol	30 30 30 30 30 31 20	000001
Leg Ratio	FF FF FF FF	-1 = Sell 1
Leg Security Type	4F	Option Leg
Leg Symbol	30 30 30 30 30 32 20	000002
Leg Ratio	01 00 00 00	1 = Buy 1
Leg Security Type	4F	Option Leg

Exchange Designated Complex Instrument Definition Message Example (C1 Only)

Table 78. Exchange Designated Complex Instrument Definition Message Example (C1 Only)

Length	51	81 bytes
Type	9F	Exchange Designated Complex Instrument Definition
Time Offset	18 D2 06 00	447,000 ns since last Time Message
CID	44 44 43 49 30 31	EDCI01
Complex Instrument Underlying	5A 56 5A 5A 54 20 20 20	ZVZZT
EDCI Type	51 53 42 20 20 20 20 20 20 20 20 20	QSB
EDCI Subtype	4A 45 4C 4C 59 5F 52 4F 4C 4C 20 20 20 20 20 20 20 20 20	JELLY_ROLL
Reserved	00 00	Reserved
Leg Count	02	2 Legs
Leg Symbol	30 30 30 30 30 31	000001
Leg Ratio	FF FF FF FF	-1 = Sell 1
Leg Symbol	30 30 30 30 30 32	000002
Leg Ratio	01 00 00 00	1 = Buy 1

Symbol Mapping Message Example

Table 79. Symbol Mapping Message Example

Length	26	38 bytes
Type	2E	Symbol Mapping Message
Feed Symbol	30 30 6D 45 56 4F	00mEVO
OSI Symbol	4D 53 46 54 20 20 31 39 30	MSFT 190920C00150000
	39 32 30 43 30 30 31 35 30	
	30 30 30	
Symbol	43	'C' - Closing
		OnlyCondition
Underlying	4D 53 46 54 20 20 20 20	MSFT

Multicast Configuration

Production Environment Configuration

Limitations/Configurations

The following table defines Cboe current configuration for network and gap request limitations. These limitations are session based. Cboe reserves the right to adjust the gap request limitations to improve the effectiveness of the gap request infrastructure.

Table 80. Limitations/Configurations

PERIOD/TYPE	LIMIT/SETTING	NOTES
MTU	1500	Cboe will send UDP messages up to 1500 bytes. Customers should ensure that their infrastructure is configured accordingly.
Gig-Shaped Throttle	1 Gb/s	The real-time and gap multicast head ends are configured to shape their output to this level to minimize packet loss.
Gap Response Delay	2 ms	The Gap Server will delay resending sequenced messages via multicast for the specified limit in order to satisfy multiple GRP gap requests with one multicast response.
Count	100	Any single gap request may not be for more than this number of dropped messages.
1 Second	320 Requests	This is the maximum number of retransmission requests allowed per second for each session. This is renewed every clock second.
1 Minute	1500 Requests	This is the maximum number of retransmission requests allowed per minute for each session. This is renewed every clock minute.
Day	100,000 Requests	This is the maximum number of retransmission requests allowed per day for each session.
Within Range	1,000,000 Messages	Users' retransmission requests must be within this many messages of the most recent sequence sent by the real-time feed.

US Options Unit/Product Distribution

UNIT	BZX/C1/C2/EDGX SYMBOL RANGE	EXCEPTIONS
1	A - AIPA~	
2	AIPB - AO~	
3	AP - AVGO~	
4	AVGP - BOIL~	
5	BOIM - CMPR~	Excludes CBTX, CBTXW
6	CMPS - CSIQ~	
7	CSIR - DJTA~	
8	DJTB - FARO~	Excludes DJX
9	FARP - GLC~	
10	GLD - HAYW~	
11	HAYX - IYWA~	Excludes IWM
12	IYWB - LMTA~	
13	LMTB - MES~	Excludes MBTX, MBTXW
14	MET - MRM~	Excludes MGTN, MGTNW
15	MRN - MSTR~	Excludes MRUT
16	MSTS - NKEA~	Excludes NANOS
17	NKEB - PDC~	Excludes NVDA, OEX
18	PDD - RDDS~	Excludes QQQ
19	RDDT - SEAA~	Excludes RLG, RLV, RUI, RUT/RUTW
20	SEAB - SMHA~	Excludes SIXB, SIXC, SIXE, SIXI, SIXR, SIXRE, SIXT, SIXU, SIXV, SIXY
21	SMHB - TEQ~	Excludes SPEQX, SPESG, SPX/SPXW, SPY
22	TER - TTC~	Excludes TSLA
23	TTD - VB~	Excludes UKXM, VIX, VIXW
24	VC - WYNN~	
25	WYNO - Z~	Excludes XEO, XSP, XSPBW*
26	NVDA	
27	TSLA	
28	QQQ	
29	IWM	
30	SPY	

Table 81. Units 31-35

UNIT	BZX/C2 SYMBOL RANGE	C1 SYMBOL RANGE
31	DJX (C2 Only), RUT (BZX and C2 Only), RUTW (C2 Only)	CBTX, CBTXW, DJX, MBTX, MBTXW, MGTN, MGTNW, MRUT, OEX, RLG, RLV, RUI, RUT, RUTW, SIXB, SIXC, SIXE, SIXI, SIXR, SIXRE, SIXT, SIXU, SIXV, SIXY, SPEQX, SPESG, UKXM, XEO
32	N/A	NANOS, VIX, VIXW, XSP, XSPBW*
33	N/A	SPX
34	N/A	SPXW
35	N/A	SPX/SPXW, Cross Product Spreads

Note - Cboe reserves the right to add units and/or change symbol distribution with 48 hours of notice and no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

***Effective 06/01/26**

BZX Options Multicast Routing Parameters

Table 82. BZX Options Multicast Routing Parameters - Production

DATA CENTER	RENDEZVOUS POINT
Primary Data Center A feed	74.115.128.33
Primary Data Center B feed	74.115.128.36
Secondary Data Center E feed	174.136.181.245

C1 Options Multicast Routing Parameters

Table 83. C1 Options Multicast Routing Parameters - Production

DATA CENTER	RENDEZVOUS POINT
Primary Data Center A feed	74.115.128.183
Primary Data Center B feed	74.115.128.184
Secondary Data Center E feed	174.136.181.249

C2 Options Multicast Routing Parameters

Table 84. C2 Options Multicast Routing Parameters - Production

DATA CENTER	RENDEZVOUS POINT
Primary Data Center A feed	74.115.128.176
Primary Data Center B feed	74.115.128.177
Secondary Data Center E feed	170.137.16.134

EDGX Options Multicast Routing Parameters

Table 85. EDGX Options Multicast Routing Parameters - Production

DATA CENTER	RENDEZVOUS POINT
Primary Data Center A feed	74.115.128.162
Primary Data Center B feed	74.115.128.163
Secondary Data Center E feed	174.136.181.240

For additional information about physical connectivity, refer to the [US Equities/Options Connectivity Manual](#).

BZX Options Address/Unit Distribution

The following tables describe the unit distribution across the BZX Complex Multicast PITCH feeds.

Table 86. BZX Options Data Center - Primary

PRIMARY DATACENTER		GIG-SHAPED [OAC] 174.136.164.192/28		GIG-SHAPED [OBC] 174.136.164.208/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESP. MC	REAL-TIME MC	GAP RESP. MC
1	30801	224.0.73.232	224.0.73.234	233.65.120.24	233.65.120.26
2	30802				
3	30803				
4	30804				
5	30805				
6	30806				
7	30807				
8	30808				
9	30809				
10	30810				
11	30811				
12	30812				
13	30813				
14	30814				
15	30815				
16	30816				
17	30817	224.0.73.233	224.0.73.235	233.65.120.25	233.65.120.27
18	30818				
19	30819				
20	30820				
21	30821				
22	30822				
23	30823				
24	30824				
25	30825				
26	30826				
27	30827				
28	30828				
29	30829				
30	30830				
31	30831				
32	30832				
33	30833				

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration. Addresses in the gray area are pre-assigned but not available. Customers should not configure their networks or systems for these addresses.

Table 87. BZX Options Datacenter - Secondary

SECONDARY DATACENTER		GIG-SHAPED [OEC] 174.136.181.128/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESPONSE MC
1	31301	234.170.137.24	234.170.137.26
2	31302		
3	31303		
4	31304		
5	31305		
6	31306		
7	31307		
8	31308		
9	31309		
10	31310		
11	31311		
12	31312		
13	31313		
14	31314		
15	31315		
16	31316		
17	31317	234.170.137.25	234.170.137.27
18	31318		
19	31319		
20	31320		
21	31321		
22	31322		
23	31323		
24	31324		
25	31325		
26	31326		
27	31327		
28	31328		
29	31329		
30	31330		
31	31331		
32	31332		
33	31333		

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

C1 Options Address/Unit Distribution

The following tables describe the unit distribution across the C1 Complex Multicast PITCH feeds.

Table 88. C1 Options Primary Data Center - Production

PRIMARY DATACENTER		GIG-SHAPED [CAC] 170.137.114.80/28		GIG-SHAPED [CBC] 170.137.115.80/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESP. MC	REAL-TIME MC	GAP RESP. MC
1	30351	224.0.74.80	224.0.74.82	233.182.199.208	233.182.199.210
2	30352				
3	30353				
4	30354				
5	30355				
6	30356				
7	30357				
8	30358				
9	30359				
10	30360				
11	30361				
12	30362				
13	30363				
14	30364				
15	30365				
16	30366				
17	30367	224.0.74.81	224.0.74.83	233.182.199.209	233.182.199.211
18	30368				
19	30369				
20	30370				
21	30371				
22	30372				
23	30373				
24	30374				
25	30375				
26	30376				
27	30377				
28	30378				
29	30379				
30	30380				
31	30381				
32	30382				
33	30383				
34	30384				
35	30385				

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration. Addresses in the gray area are pre-

assigned but not available. Customers should not configure their networks or systems for these addresses.

Table 89. C1 Options Secondary Datacenter - Production

SECONDARY DATACENTER		GIG-SHAPED [CEC] 170.137.124.224/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESPONSE MC
1	31351	233.19.3.240	233.19.3.242
2	31352		
3	31353		
4	31354		
5	31355		
6	31356		
7	31357		
8	31358		
9	31359		
10	31360		
11	31361		
12	31362		
13	31363		
14	31364		
15	31365		
16	31366		
17	31367	233.19.3.241	233.19.3.243
18	31368		
19	31369		
20	31370		
21	31371		
22	31372		
23	31373		
24	31374		
25	31375		
26	31376		
27	31377		
28	31378		
29	31379		
30	31380		
31	31381		
32	31382		
33	31383		
34	31384		
35	31385		

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

C2 Options Address/Unit Distribution

The following tables describe the unit distribution across the C2 Complex Multicast PITCH feeds.

Table 90. C2 Options Data Center - Primary

PRIMARY DATACENTER		GIG-SHAPED [WAC] 174.136.164.64/28		GIG-SHAPED [WBC] 174.136.164.80/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESP. MC	REAL-TIME MC	GAP RESP. MC
1	30301	224.0.131.248	224.0.131.250	233.130.124.248	233.130.124.250
2	30302				
3	30303				
4	30304				
5	30305				
6	30306				
7	30307				
8	30308				
9	30309				
10	30310				
11	30311				
12	30312				
13	30313				
14	30314				
15	30315				
16	30316				
17	30317	224.0.131.249	224.0.131.251	233.130.124.249	233.130.124.251
18	30318				
19	30319				
20	30320				
21	30321				
22	30322				
23	30323				
24	30324				
25	30325				
26	30326				
27	30327				
28	30328				
29	30329				
30	30330				
31	30331				
32	30332				
33	30333				

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration. Addresses in the gray area are pre-assigned but not available. Customers should not configure their networks or systems for these addresses.

Table 91. C2 Options Datacenter - Secondary

SECONDARY DATACENTER		GIG-SHAPED [WEC] 170.137.17.96/29	
UNIT	IP PORT	REAL-TIME MC	GAP RESPONSE MC
1	31301	233.182.199.104	233.182.199.106
2	31302		
3	31303		
4	31304		
5	31305		
6	31306		
7	31307		
8	31308		
9	31309		
10	31310		
11	31311		
12	31312		
13	31313		
14	31314		
15	31315		
16	31316		
17	31317	233.182.199.105	233.182.199.107
18	31318		
19	31319		
20	31320		
21	31321		
22	31322		
23	31323		
24	31324		
25	31325		
26	31326		
27	31327		
28	31328		
29	31329		
30	31330		
31	31331		
32	31332		
33	31333		

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

EDGX Options Address/Unit Distribution

The following tables describe the unit distribution across the EDGX Complex Multicast PITCH feeds.

Table 92. EDGX Options Datacenter - Primary

PRIMARY DATACENTER		GIG-SHAPED [EAC] 174.136.164.32/28		GIG-SHAPED [EBC] 174.136.164.48/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESP. MC	REAL-TIME MC	GAP RESP. MC
1	30551	224.0.131.152	224.0.131.154	233.130.124.152	233.130.124.154
2	30552				
3	30553				
4	30554				
5	30555				
6	30556				
7	30557				
8	30558				
9	30559				
10	30560				
11	30561				
12	30562				
13	30563				
14	30564				
15	30565				
16	30566				
17	30567	224.0.131.153	224.0.131.155	233.130.124.153	233.130.124.155
18	30568				
19	30569				
20	30570				
21	30571				
22	30572				
23	30573				
24	30574				
25	30575				
26	30576				
27	30577				
28	30578				
29	30579				
30	30580				
31	30581				
32	30582				
33	30583				

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration. Addresses in the gray area are pre-assigned but not available. Customers should not configure their networks or systems for these addresses.

Table 93. EDGX Options Datacenter - Secondary

SECONDARY DATACENTER		GIG-SHAPED [EEC] 174.136.176.144/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESPONSE MC
1	31951	233.19.3.136	233.19.3.138
2	31952		
3	31953		
4	31954		
5	31955		
6	31956		
7	31957		
8	31958		
9	31959		
10	31960		
11	31961		
12	31962		
13	31963		
14	31964		
15	31965		
16	31966		
17	31967	233.19.3.137	233.19.3.139
18	31968		
19	31969		
20	31970		
21	31971		
22	31972		
23	31973		
24	31974		
25	31975		
26	31976		
27	31977		
28	31978		
29	31979		
30	31980		
31	31981		
32	31982		
33	31983		

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

Certification Environment Configuration

US Options Unit/Product Distribution

UNIT	BZX/C1/C2/EDGX SYMBOL RANGE	EXCEPTIONS
1	A - AIPA~	
2	AIPB - AO~	
3	AP - AVGO~	
4	AVGP - BOIL~	
5	BOIM - CMPR~	Excludes CBTX, CBTXW
6	CMPS - CSIQ~	
7	CSIR - DJTA~	
8	DJTB - FARO~	Excludes DJX
9	FARP - GLC~	
10	GLD - HAYW~	
11	HAYX - IYWA~	Excludes IWM
12	IYWB - LMTA~	
13	LMTB - MES~	Excludes MBTX, MBTXW
14	MET - MRM~	Excludes MGTN, MGTNW
15	MRN - MSTR~	Excludes MRUT
16	MSTS - NKEA~	Excludes NANOS
17	NKEB - PDC~	Excludes NVDA, OEX
18	PDD - RDDS~	Excludes QQQ
19	RDDT - SEAA~	Excludes RLG, RLV, RUI, RUT/RUTW
20	SEAB - SMHA~	Excludes SIXB, SIXC, SIXE, SIXI, SIXR, SIXRE, SIXT, SIXU, SIXV, SIXY
21	SMHB - TEQ~	Excludes SPEQX, SPESG, SPX/SPXW, SPY
22	TER - TTC~	Excludes TSLA
23	TTD - VB~	Excludes UKXM, VIX, VIXW
24	VC - WYNN~	
25	WYNO - Z~	Excludes XEO, XSP, XSPBW*
26	NVDA	
27	TSLA	
28	QQQ	
29	IWM	
30	SPY	

Table 94. Units 31-35

UNIT	BZX/C2 SYMBOL RANGE	C1 SYMBOL RANGE
31	DJX (C2 Only), RUT (BZX and C2 Only), RUTW (C2 Only)	CBTX, CBTXW, DJX, MBTX, MBTXW, MGTN, MGTNW, MRUT, OEX, RLG, RLV, RUI, RUT, RUTW, SIXB, SIXC, SIXE, SIXI, SIXR, SIXRE, SIXT, SIXU, SIXV, SIXY, SPEQX, SPESG, UKXM, XEO
32	N/A	NANOS, VIX, VIXW, XSP, XSPBW*
33	N/A	SPX
34	N/A	SPXW
35	N/A	SPX/SPXW, Cross Product Spreads

Note - Cboe reserves the right to add units and/or change symbol distribution with 48 hours of notice and no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

***Effective 06/01/26**

Multicast Routing Parameters

PRIMARY CERTIFICATION DATA CENTER	RENDEZVOUS POINT
C1	74.115.128.131
C2, EDGX, and BZX	74.115.128.129

BZX Options Address/Unit Distribution

The following tables describe the unit distribution across certification BZX Options Complex Multicast PITCH feeds out of the Primary datacenter.

PRIMARY DATACENTER		CERTIFICATION 174.136.164.224/28			
UNIT	IP PORT	REAL-TIME MC	GAP RESP. MC	REAL-TIME MC	GAP RESP. MC
1	32551	224.0.62.232	224.0.62.234		
2	32552				
3	32553				
4	32554				
5	32555				
6	32556				
7	32557				
8	32558				
9	32559				
10	32560				
11	32561				
12	32562				
13	32563				
14	32564				
15	32565				
16	32566				
17	32567	224.0.62.233	224.0.62.235		
18	32568				
19	32569				
20	32570				
21	32571				
22	32572				
23	32573				
24	32574				
25	32575				
26	32576				
27	32577				
28	32578				
29	32579				
30	32580				
31	32581				
32	32582				
33	32583				

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

C1 Options Address/Unit Distribution

The following table describes the unit distribution across certification C1 Complex Multicast PITCH feeds out of the Primary datacenter.

Table 95. C1 Options Address/Unit Distribution - Certification

PRIMARY DATACENTER		CERTIFICATION 170.137.126.16/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESP. MC
1	32351	233.103.126.8	233.103.126.10
2	32352		
3	32353		
4	32354		
5	32355		
6	32356		
7	32357		
8	32358		
9	32359		
10	32360		
11	32361		
12	32362		
13	32363		
14	32364		
15	32365		
16	32366		
17	32367	233.103.126.9	233.103.126.11
18	32368		
19	32369		
20	32370		
21	32371		
22	32372		
23	32373		
24	32374		
25	32375		
26	32376		
27	32377		
28	32378		
29	32379		
30	32380		
31	32381		
32	32382		
33	32383		
34	32384		
35	32385		

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

C2 Options Address/Unit Distribution

The following table describes the unit distribution across certification C2 Complex Multicast PITCH feeds out of the Primary datacenter.

Table 96. C2 Options Address/Unit Distribution - Certification

PRIMARY DATACENTER		CERTIFICATION 174.136.160.80/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESP. MC
1	32251	224.0.74.160	224.0.74.162
2	32252		
3	32253		
4	32254		
5	32255		
6	32256		
7	32257		
8	32258		
9	32259		
10	32260		
11	32261		
12	32262		
13	32263		
14	32264		
15	32265		
16	32266		
17	32267	224.0.74.161	224.0.74.163
18	32268		
19	32269		
20	32270		
21	32271		
22	32272		
23	32273		
24	32274		
25	32275		
26	32276		
27	32277		
28	32278		
29	32279		
30	32280		
31	32281		
32	32282		
33	32283		

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

EDGX Options Address/Unit Distribution

The following table describes the unit distribution across certification EDGX Complex Multicast PITCH feeds out of the Primary datacenter.

Table 97. EDGX Options Address/Unit Distribution - Certification

PRIMARY DATACENTER		CERTIFICATION 174.136.174.176/28	
UNIT	IP PORT	REAL-TIME MC	GAP RESP. MC
1	32551	224.0.74.184	224.0.74.186
2	32552		
3	32553		
4	32554		
5	32555		
6	32556		
7	32557		
8	32558		
9	32559		
10	32560		
11	32561		
12	32562		
13	32563		
14	32564		
15	32565		
16	32566		
17	32567	224.0.74.185	224.0.74.187
18	32568		
19	32569		
20	32570		
21	32571		
22	32572		
23	32573		
24	32574		
25	32575		
26	32576		
27	32577		
28	32578		
29	32579		
30	32580		
31	32581		
32	32582		
33	32583		

Note - Cboe reserves the right to add multicast addresses with prior notice, but no migration period. Notice will be given that the distribution will change on a certain date. Care should be taken to support mappings in these tables via software configuration.

Options Trade Condition Codes

The following table defines valid values for the *Trade Condition* field.

Table 98. Options *Trade Condition* Codes

TYPE	FIELD VALUE
f	Complex to Complex Electronic Trade. Cboe auction type is COA
g	Complex Auction Trade. Cboe order types include C-AIM, C-SAM
h	Complex Cross. Cboe auction types include Cust to Cust C-AIM, C-QCC
j	Complex Electronic Trade Against Single Leg(s)
k	Complex with Stock Options Auction Trade. Cboe auction types include C-AIM w/ Stock, C-SAM w/ Stock
m	Complex Floor Trade Against Single Leg(s). All complex floor executions are reported as condition 'm'.
n	Complex with Stock Electronic Trade. Includes COA auctions done electronically
o	Complex with Stock Cross. Cboe auction types include C-QCC w/ Stock
p	Complex with Stock Floor Trade
t	Complex Floor Trade of Proprietary Products Marked as "Combo Order"
v	Extended Hours Trade. Transaction represents a trade executed during the Curb session.
l	Electronic Trade
O*	Opening Trade

*The *Trade Condition* value of O=Opening Trade will continue to be disseminated on the options PITCH and TOP feeds but will not be sent to OPRA.

Connectivity

Supported Extranet Carriers

Cboe has certified a number of carriers defined in the [Cboe US Equity/Options Connectivity Manual](#) with respect to redistribution of Cboe Multicast data feeds. For more information on receiving Multicast PITCH through any of these providers, reach out to the vendor contact noted in the [Cboe US Equity/Options Connectivity Manual](#) under Extranet Providers.

Bandwidth Recommendation

The Gig-shaped feeds require 1Gbps of bandwidth. Cboe will use 90% of these respective bandwidths for Multicast PITCH to allow customers to use the same physical connection for FIX order entry if desired.

Multicast Test Program

The ZIP file located at https://cdn.cboe.com/resources/membership/mcast_pitch.zip contains a sample program that may be used to test Multicast PITCH feed connections and to troubleshoot Multicast issues. Refer to the included README file for build and usage information.

References

For more information on Cboe Symbology, please see [Cboe Symbology Reference](#) document.

Support

Please direct questions or comments regarding this specification to tradedesk@cboe.com.

Revision History

DOCUMENT VERSION	DATE	DESCRIPTION
2.0.0	05/11/17	Initial version of US Options Complex Multicast PITCH Specification based on Multicast PITCH 2.X.
2.0.1	05/18/17	Various minor updates and clarification added.
2.0.2	07/28/17	Added Multicast Ips/Ports for Certification environment. Added Auction Update and Auction Summary messages.
2.0.3	08/08/17	Added Multicast Ips/Ports for Production environment.
2.0.4	09/01/17	Added C2 Options references.
2.0.5	10/17/17	Cboe branding/logo changes.
2.0.6	11/24/17	Added C2 Options Certification IP and Port information. Added RUT, RUTW options (C2 Options Only) to distinct unit (unit 33).
2.0.7	02/05/18	Removed the "A" <i>Trading Status</i> field value as this is used for equities only. Added C2 Options Production IP and Port information.
2.0.8	03/08/18	Updated Unit Distribution ranges.
2.0.9	03/23/18	Unit Distribution ranges Effective Date updated to 4/14/18.
2.1.0	11/16/18	Added support for Cboe Options Exchange.
2.1.1	12/06/18	Added notes identifying Feature Pack 4 updates.
2.1.2	12/21/19	Removed Floor Trade value from <i>Trade Condition</i> field, as this was added in error. Added a note of clarification, indicating that a Trade message can also be sent when an auction executes against a non-displayed order, such as a contra response.
2.1.3	02/14/19	Corrected value of Complex AIM value to "B" for <i>Auction Type</i> field in Auction Notification message. Added certification IP port information.
2.1.4	03/04/19	Added matching engine unit 33 information in support of XSP trading on EDGX Options effective 04/08/19. Added C1 primary data center rendezvous point IP address and C1 Certification symbol ranges.
2.1.5	04/15/19	Added C1 production IP port and unit distribution. Transaction Begin and Transaction End messages are currently restricted to C1 only. Added DJX to C2 ME 33 in Unit/Product Distribution tables (effective 05/08/19).
2.1.6	05/01/19	Added note indicating Transaction Begin and Transaction End messages will be disseminated for C2 and EDGX options (effective with C1 Feature Pack 7).
2.1.7	05/08/19	Removed <i>Trading Status</i> value S = Exchange Specific Suspension. Corrected C1 Production Gig-Shaped [CAC] and [CBC] source network IP addresses.
2.1.8	05/14/19	Added <i>Composite Market Bid Price</i> and <i>Composite Market Offer Price</i> fields to the Options Auction Update message and updated associated example message. Added additional proprietary products to matching unit 31 in C1.
2.1.9	06/12/19	Corrected certification and production C1 symbol range for units 9 and 20.
2.1.10	08/02/19	Added note indicating Options Auction Update message <i>Opening Condition</i> field value will always be zero. Updated example message. Removed Complex Instrument Definition from list of PITCH 2.X messages.

DOCUMENT VERSION	DATE	DESCRIPTION
		Corrected Leg Count field description in Complex Instrument Definition Expanded message to indicate a total of 12 legs are allowed.
2.1.11	09/18/19	Corrected OSI Symbol example values in Symbol Mapping message type example.
2.1.12	10/03/19	Corrected UKXM symbol exclusion in Unit Distribution table. Changed instances of Complex Instrument Definition to Complex Instrument Definition Expanded , as the former was deprecated 02/28/19.
2.1.13	10/31/19	Clarified description of Time message. Added Options Trade Condition section (effective 01/13/20).
2.1.14	11/12/19	Added note indicating Unit Clear message is sent at the beginning of the day for Equities only. Added note indicating GTH will be applicable for C1 only as GTH is being sunset for C2 and EDGX (effective 11/22/19).
2.1.15	12/19/19	Updated Options Trade Condition Codes by adding 'O' =Opening Trade and correcting field value description for 'p'by removing "Includes Complex Auctions on the Floor". (Effective 01/13/20).
2.1.16	01/03/20	Updated description of Options Trade Condition Code 't' to read, Complex Floor Trade of Proprietary Products Marked as "Combo Order" .
2.1.17	01/08/20	Removed I = Complex Auction Against Single Legs(s) from Options Trade Condition Codes table.
2.1.18	01/31/20	Corrected Unit Symbol Distribution tables to indicate QQQ is an exception for C1 Unit 20 as it has a dedicated location on Unit 28. Updated Complex Trade Condition Values.
2.1.19	08/27/20	Added SPESG to the Unit Symbol Distribution tables for C1 unit 31 (effective 09/21/20).
2.1.20	10/06/20	Added SPESG to the Unit Symbols Distribution Exceptions entries.
2.1.21	10/20/20	Added XSP to the Unit Symbol Distribution tables for BZX and removed it from EDGX (effective 11/2/20).
2.1.22	01/22/21	Updated Price field description on Auction Notification message to indicate that for SPX and SPXW AIM, this field will reflect the auction start price (C1 Only) (effective 02/22/21).
2.1.23	02/01/21	Added MRUT to Unit/Product Distribution tables for C1 unit 31 (effective 03/01/21). Added new updated Unit/Product Distribution tables with harmonized symbol ranges (effective 03/22/21).
2.1.24	03/03/21	Updated the Delete Order message description.
2.1.25	03/11/21	Updated the Unit Symbols Distribution Exceptions entries (effective 3/22/21).
2.1.26	03/25/21	Added Binary Date field type to Section 2.2 - Data Types (effective 10/10/21 TBD 09/27/21 Q3 2021). Added new Time Reference message (effective 10/10/21 TBD 09/27/21 Q3 2021). Added EpochTime field to Time message (effective 10/10/21 TBD 09/27/21 Q3 2021). Updated description of Auction Type field on Options Auction Update and Auction Summary messages (effective TBD 09/27/21 Q3 2021). Updated description of GTH Trading Status field on Trading Status message (effective 01/24/22 TBD 09/27/21 Q3 2021).

DOCUMENT VERSION	DATE	DESCRIPTION
2.1.27	05/13/21	Added clarification to the description of heartbeat messages indicating that when the system fails to receive two consecutive heartbeat messages within the specified interval the client connection is terminated. Updated Curb session effective date to 02/07/22 TBD 09/27/21. Added v = Extended Hours Trade Trade Condition code (effective 01/24/22 TBD 09/27/21).
2.1.28	06/08/21	Noted in Complex Instrument Definition Expanded message that 16 legs are allowed (effective 08/25/21 08/09/21).
2.1.29	06/15/21	Updated effective date for expanded GTH session to 11/21/21.
2.1.30	08/02/21	Updated the effective date for the Complex Instrument Definition Expanded message that a total of 16 legs are allowed (effective 08/25/21).
2.1.31	08/27/21	Updated Curb session and Trade Condition code v = Extended Hours effective dates to 01/23/22 TBD.
2.1.32	09/09/21	Added <i>Trading Status</i> field value L = Curb Trading (C1 Only) for Trading Status message (effective 01/24/22 TBD). <i>GTH Trading Status</i> field will not be used for Curb session. Updated description of Auction Type field on Options Auction Update and Auction Summary messages (effective TBD).
2.1.33	09/29/21	Updated effective date for new Time Reference message (C1 Only), <i>EpochTime</i> field to Time message (C1 Only), and Binary Date field type to Section 2.2 - Data Types to 10/10/21. Added new section 1.3 - '24x5 Feed Hours and System Restart (C1 Only)' (effective 10/10/21).
2.1.34	11/04/21	Corrected example Time message values. Updated Curb session effective date to 02/07/22. Updated effective date for v = Extended Hours Trade Trade Condition code to 01/24/22. Updated effective date for <i>Trading Status</i> field value 'L = Curb Trading' to 01/24/22. Removed note indicating AuctionType value O will be sent prior to Curb session. This value will only be sent for the RTH Opening.
2.1.35	01/03/22	Added m = Complex Floor Trade Against Single Leg(s), p = Complex with Stock Floor Trade, and " t = Complex Floor Trade of Proprietary Products Marked as 'Combo Order'" Trade Condition Codes. Updated Delete Order message description. Updated the Multicast Test Program ZIP file link.
2.1.36	02/02/22	Added NANOS to the C1 unit 32 Unit/Product Distribution tables (effective 03/14/22).
2.1.37	03/01/22	Removed XSP from the BZX unit 31 Unit/Product Distribution tables.
2.1.38	10/07/22	Updated EDGX Options Address/Unit Distribution Gig-Shaped (EEC).
2.1.39	11/07/22	Moved XSP to the C1 unit 32 Unit/Production Distribution tables (effective 12/04/22).
2.1.40	03/30/23	Clarified RUT is on BZX and C2 Unit 31.
2.1.41	01/29/24	Added MXACW, MXUSA, and MXWLD to the C1 unit 31 Unit/Product Distribution tables (effective 03/18/24).
2.1.42	04/02/24	Added Exchange Designated Complex Instrument Definition message (effective 06/24/24).
2.1.43	04/08/24	Added additional detail to Exchange Designated Complex Instrument Definition message (effective 06/24/24).

DOCUMENT VERSION	DATE	DESCRIPTION
2.1.44	04/24/24	Clarified Exchange Designated Complex Instrument Definition message is C1 only (effective 06/24/24).
2.1.45	06/07/24	Added a two-byte Reserve field to the Exchange Designated Complex Instrument Definition example message.
2.1.46	07/22/24	Effective 08/26/24, GTH will be extended until 9:25 a.m. ET (C1 only) .
2.1.47	08/16/24	Added two Reserved bytes to the Add Order Expanded message (effective 10/25/24 on BZX and 10/28/24 on C1, C2, and EDGX).
2.1.48	08/20/24	The Add Order Expanded message will add five Reserved bytes instead of two (effective 10/25/24 on BZX and 10/28/24 on C1, C2, and EDGX).
2.1.49	11/22/24	Added CBTX, CBTXW, MBTX, and MBTXW to the C1 unit 31 Unit/Product Distribution tables (effective 12/02/24).
2.1.50	12/13/24	Added new Unit/Product Distribution for Units 1-30 (effective 03/31/25).
2.1.51	01/15/25	Updated with Cboe Titanium branding.
2.1.52	03/25/25	Noted in Spin Servers on page 12: a Time message will be sent as the last message in a Spin if the last Time message sent on a Spin is older than the last received time from the internal market data producers. Added SPEQX to C1 unit 31 unit/product distribution on page 114 (effective 04/14/25).
2.1.53	07/08/25	Added Time Offset data type. Effective 09/08/25, Time Offset fields will be populated to nanosecond precision.
2.1.54	07/24/25	Specification updated to indicate complex functionality is available for all Cboe Options Exchanges, with BZX Options complex functionality effective on 10/13/25.
2.1.55	08/05/25	Updated Instrument Definition Response message to include <i>Length</i> field.
2.1.56	10/20/25	Added MGTN and MGTNW to unit 31 in US Options Unit/Product Distribution on page 114 and excluded MGTN and MGTNW from unit 14 (effective 12/08/25 TBD 11/17/25).
2.1.57	11/06/25	Updated MGTN and MGTNW effective dates to TBD.
2.1.58	11/18/25	Updated MGTN and MGTNW effective dates to 12/08/25.
2.1.59	01/12/26	Updated 24x5 Feed Hours and System Restart on page 6 to document the new time at which persisted orders are added back into the order book (effective 02/02/26).
2.1.60	01/13/26	Added C = SCOA to <i>Auction Type</i> on the Auction Notification Message (effective TBD).
2.1.61	04/01/26	Removed references to MXEA, MXEF, MXACW, MXUSA, and MXWLD. Updated <i>Time</i> description in Time and Time Reference messages. Added XSPBW to C1 unit 32 Unit/Product Distribution tables (effective 06/01/26).